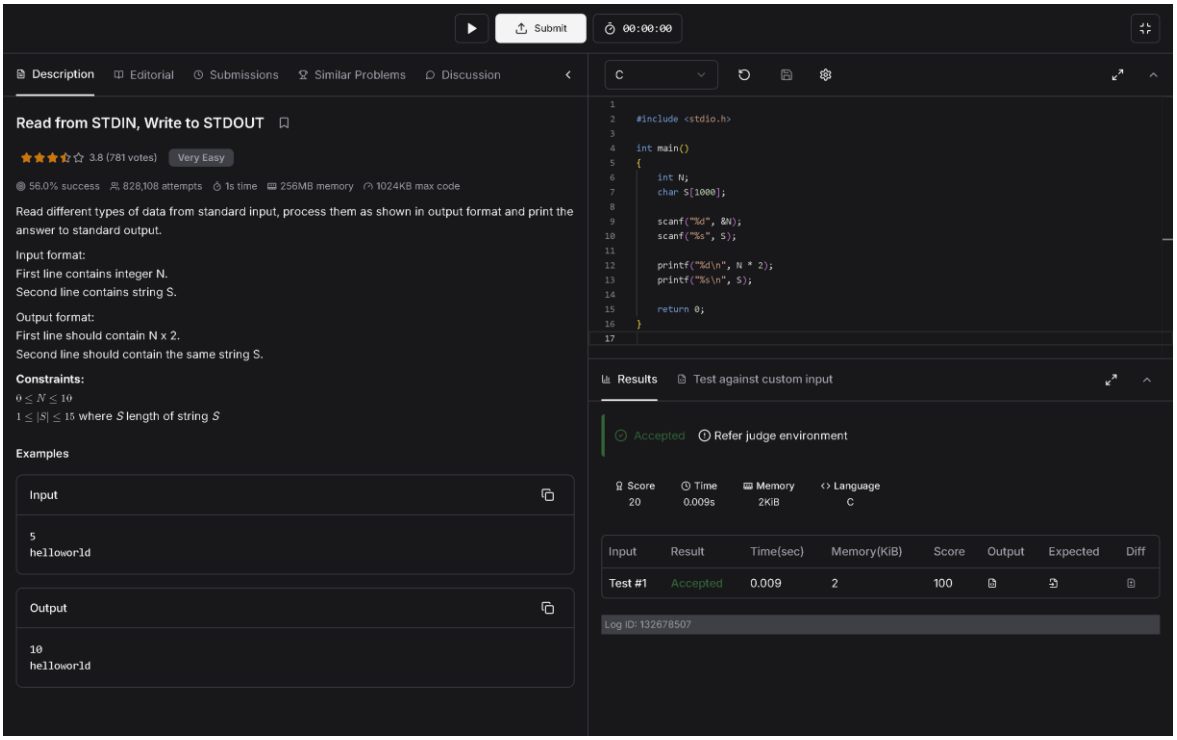
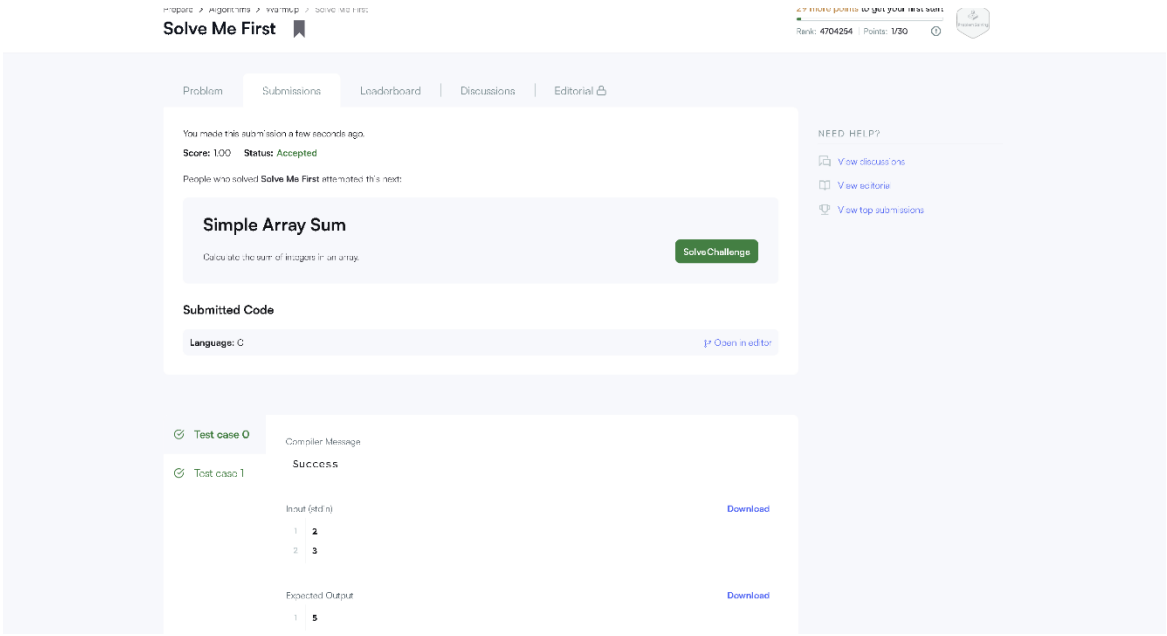
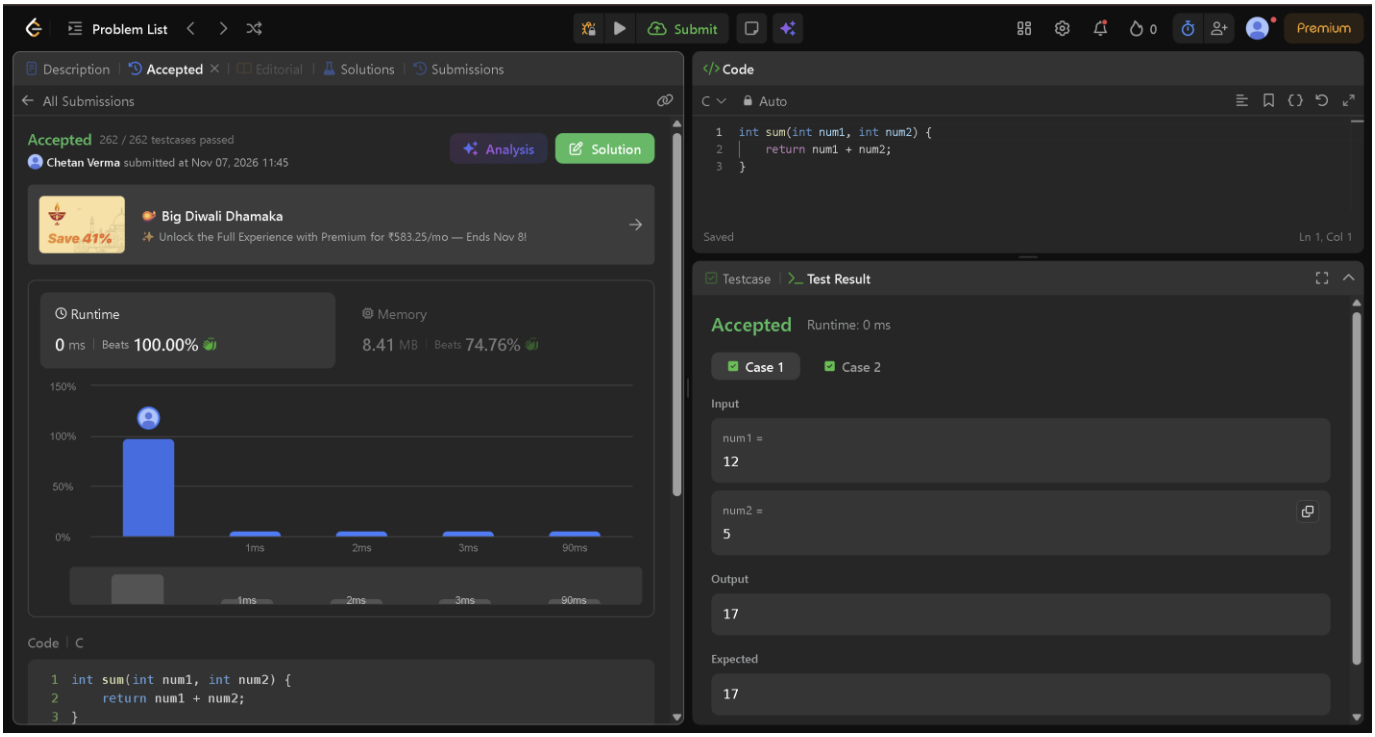
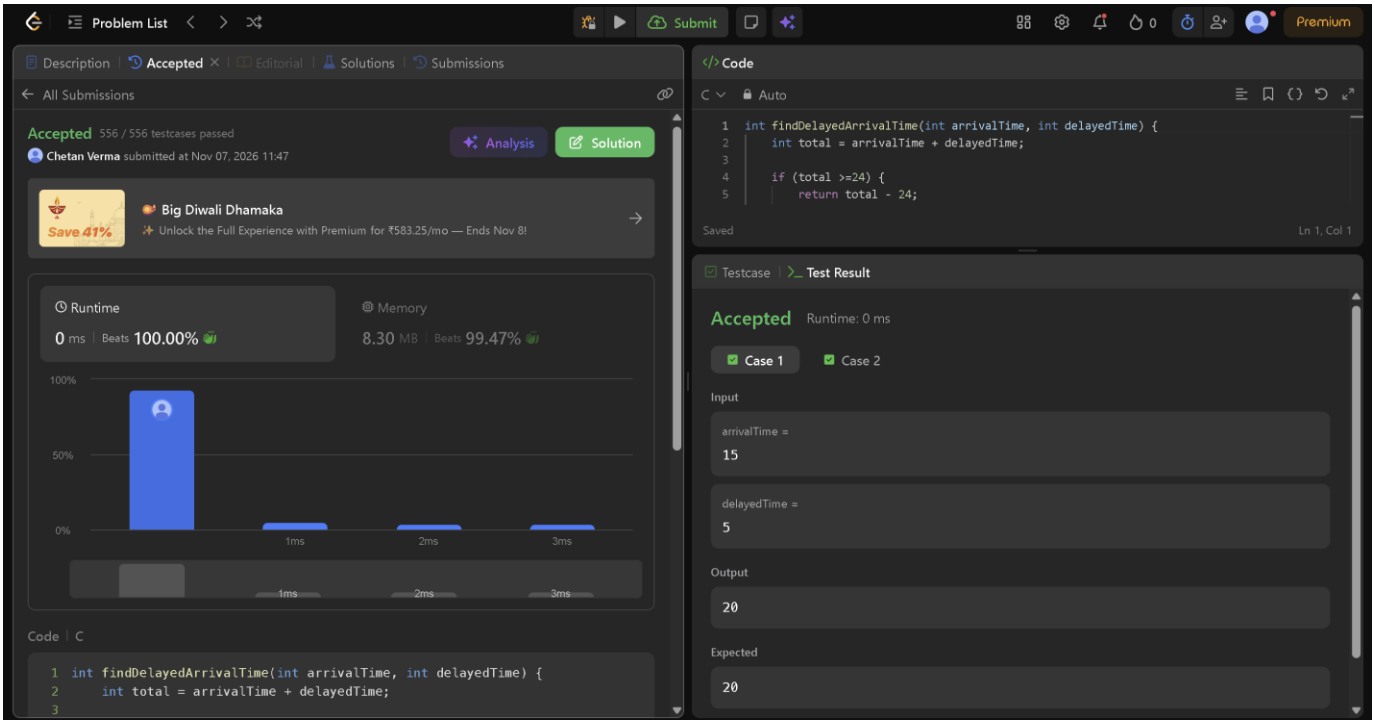
C Programming
Platform Coding Practice Question Bank

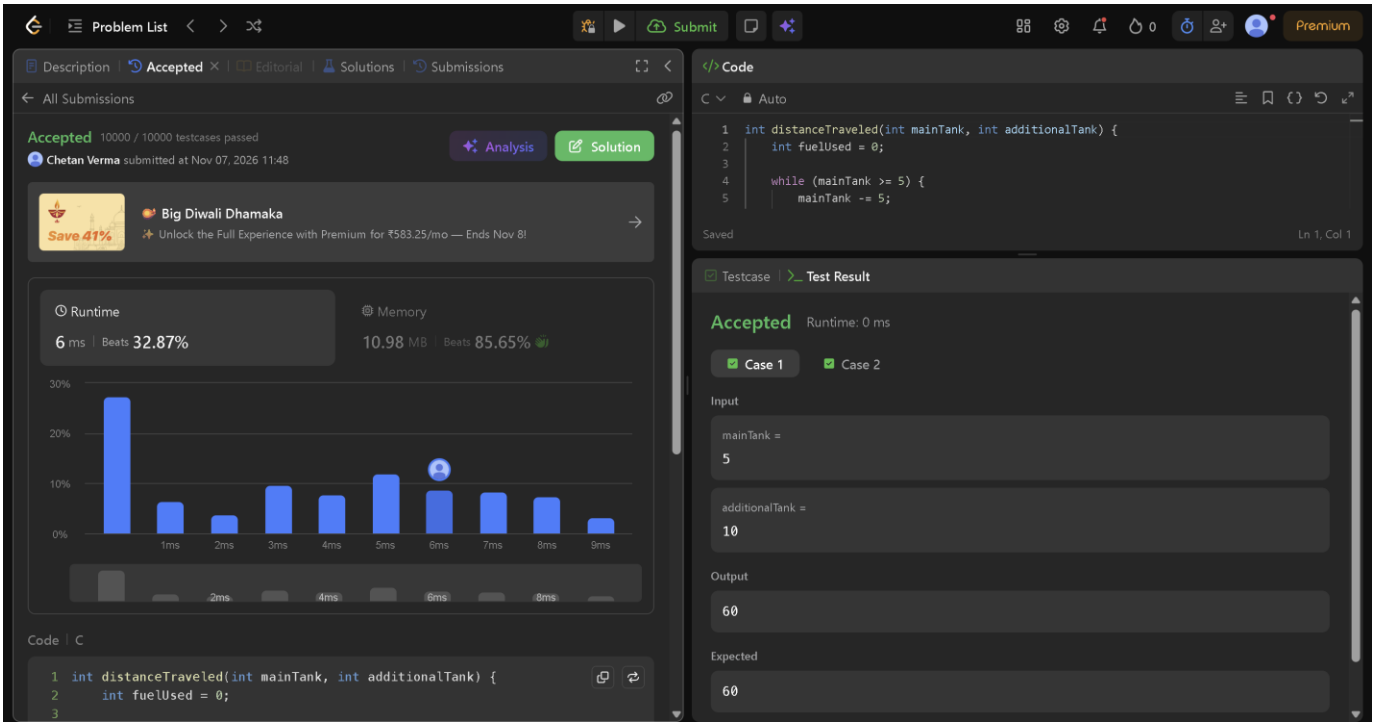
Module 1 — Input/output, variables and basic arithmetic



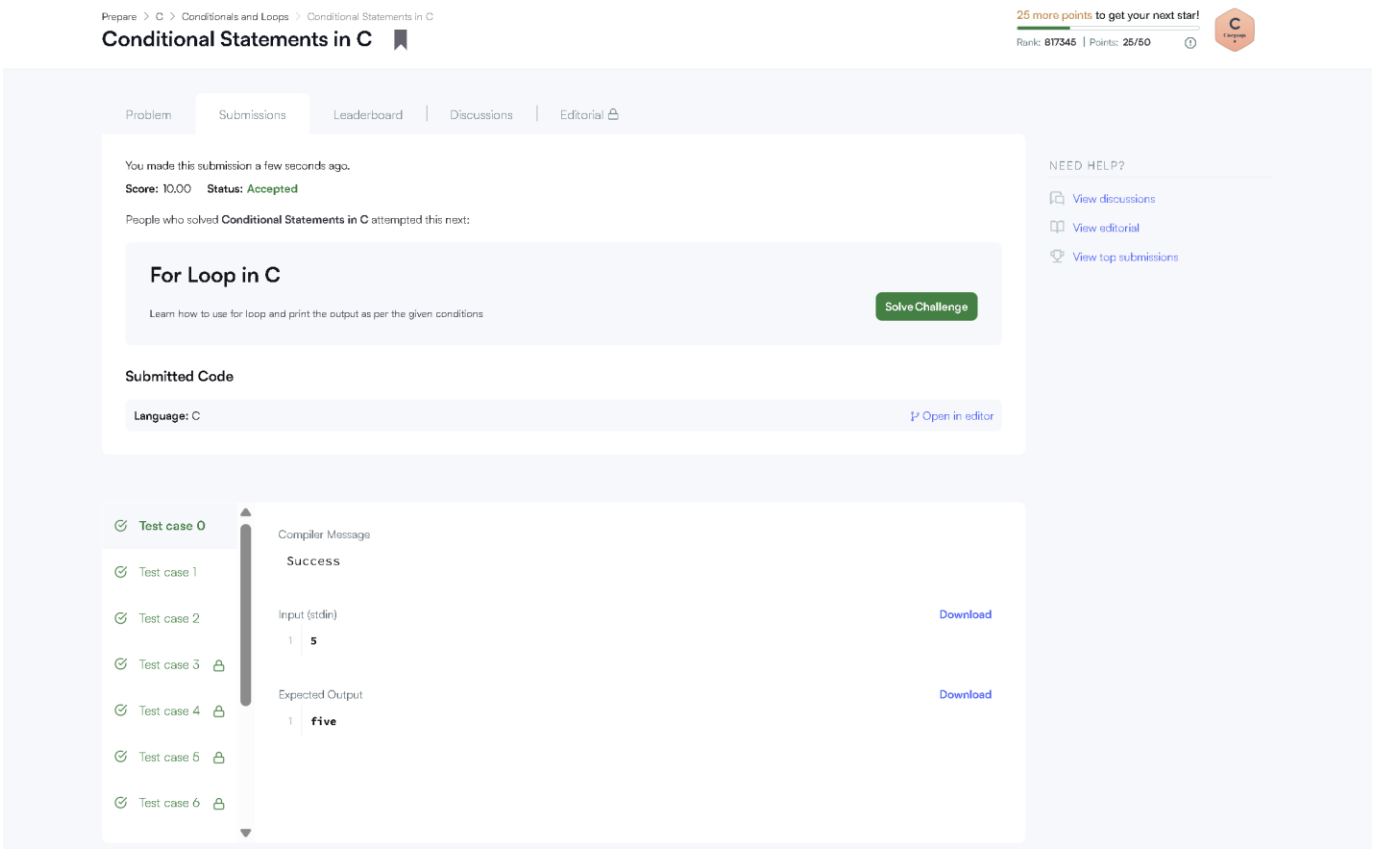


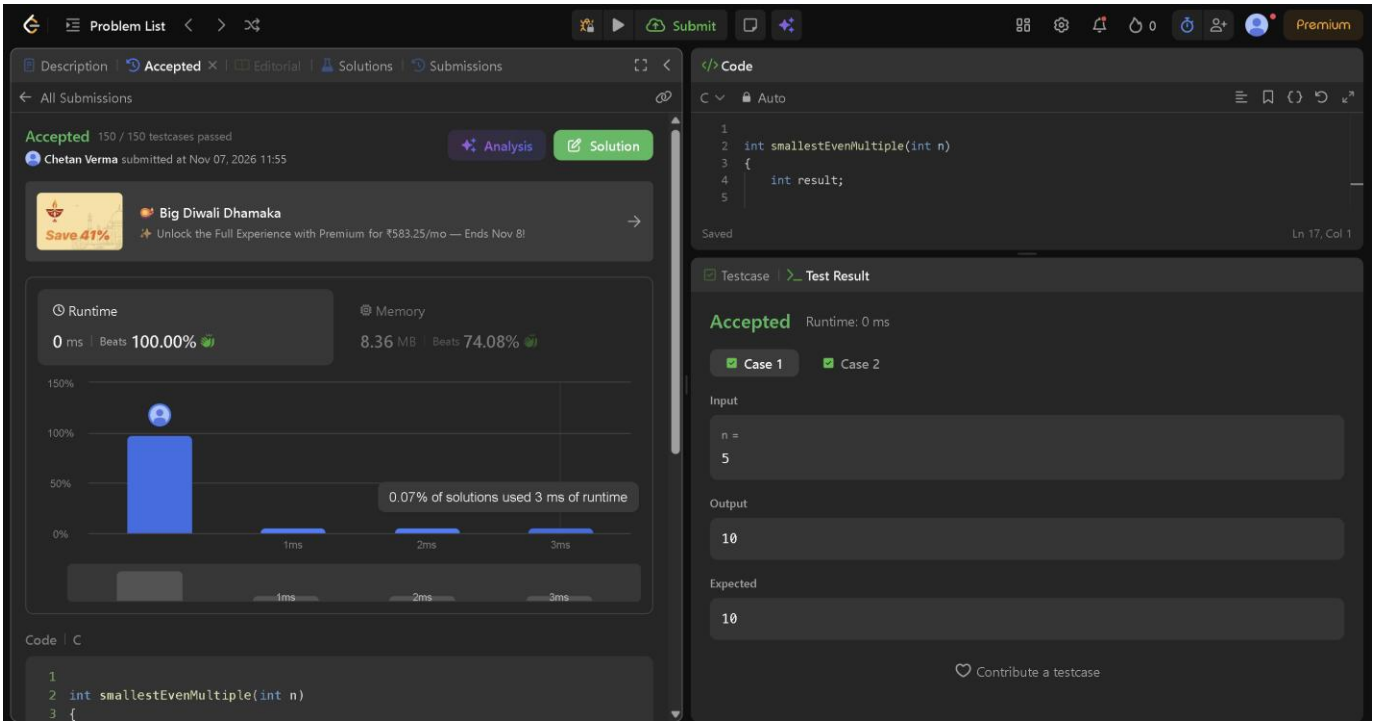

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Accepted 262 / 262 testcases passed
Chetan Verma submitted at Nov 07, 2026 11:45
Analysis
Solution
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
0 ms | Beats 100.00%
Memory
8.41 MB | Beats 74.76%
150%
100%
50%
0%
1ms
2ms
3ms
90ms
Code | C
1 int sum(int num1, int num2) {
2 return num1 + num2;
3 }
Code
C Auto
1 int sum(int num1, int num2) {
2 return num1 + num2;
3 }
Saved
Ln 1, Col 1
Testcase | Test Result
Accepted Runtime: 0 ms
Case 1
Case 2
Input
num1 =
12
num2 =
5
Output
17
Expected
17

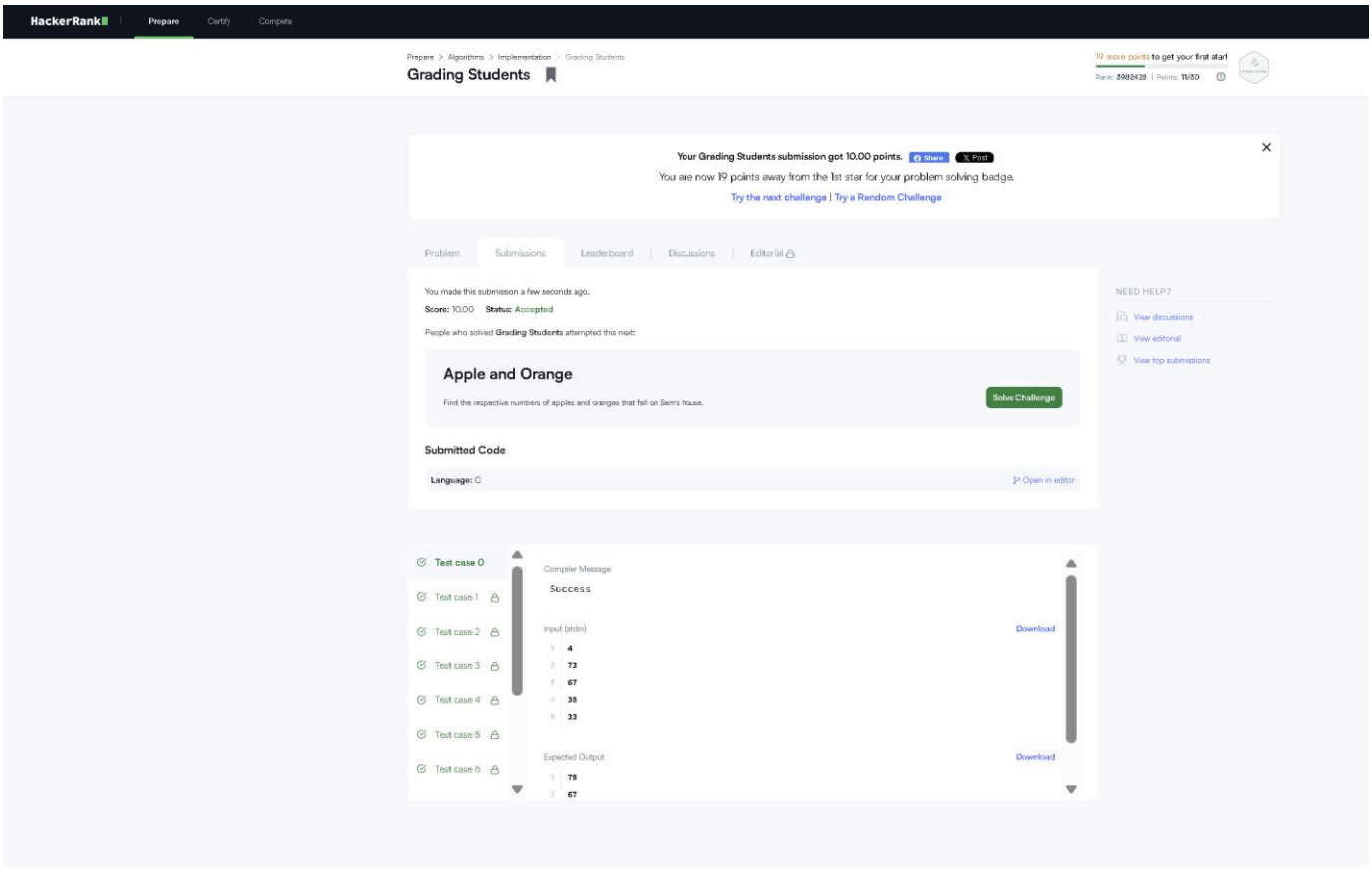

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Accepted 556 / 556 testcases passed
Chetan Verma submitted at Nov 07, 2026 11:47
Analysis
Solution
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
0 ms | Beats 100.00%
Memory
8.30 MB | Beats 99.47%
100%
50%
0%
1ms
2ms
3ms
Code | C
1 int findDelayedArrivalTime(int arrivalTime, int delayedTime) {
2 int total = arrivalTime + delayedTime;
3
Code
C Auto
1 int findDelayedArrivalTime(int arrivalTime, int delayedTime) {
2 int total = arrivalTime + delayedTime;
3
4 if (total >=24) {
5 return total - 24;
Saved
Ln 1, Col 1
Testcase | Test Result
Accepted Runtime: 0 ms
Case 1
Case 2
Input
arrivalTime =
15
delayedTime =
5
Output
20
Expected
20

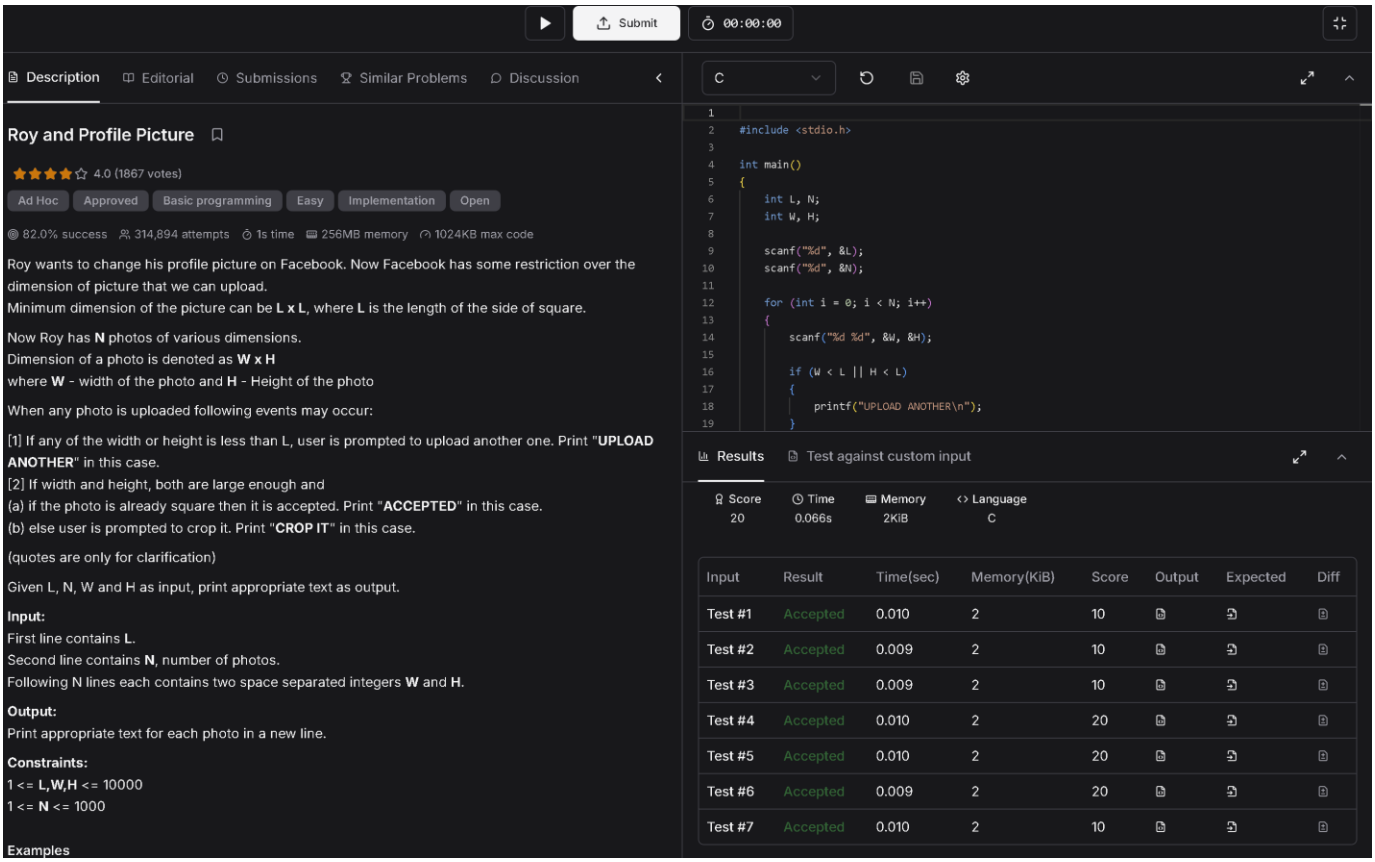


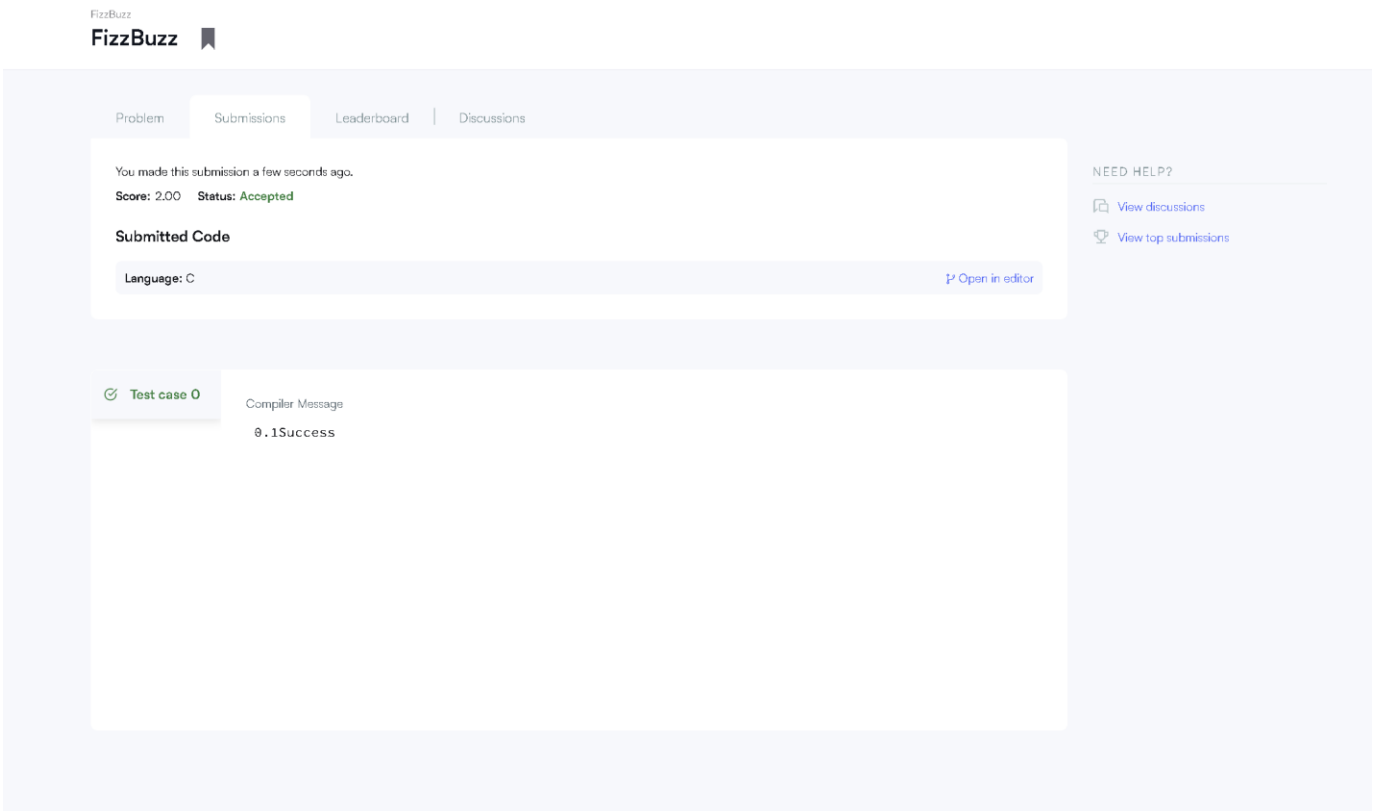
Module 2 — Conditional statements and operators



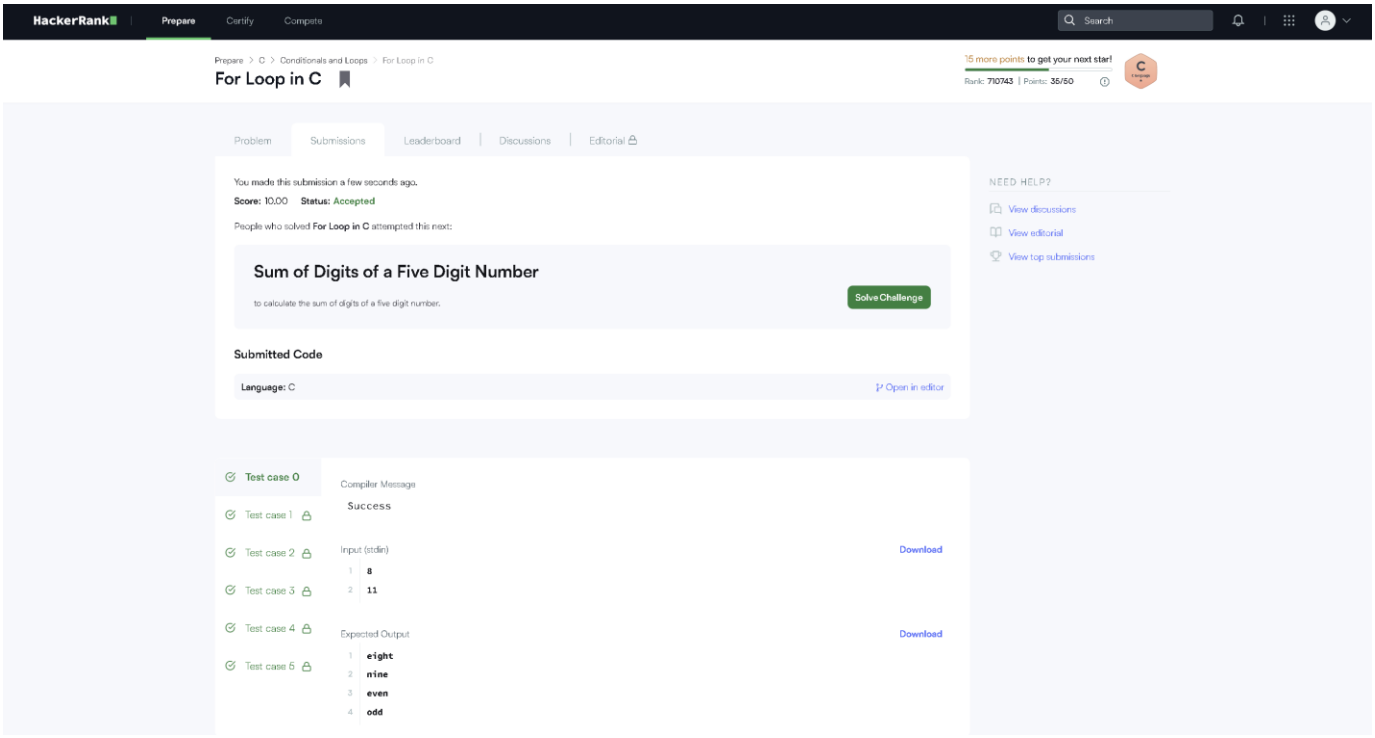

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Accepted 150 / 150 testcases passed
Chetan Verma submitted at Nov 07, 2026 11:55
Analysis
Solution
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
0 ms | Beats 100.00%
Memory
8.36 MB | Beats 74.08%
150%
100%
50%
0%
1ms
2ms
3ms
0.07% of solutions used 3 ms of runtime
Code | C
int smallestEvenMultiple(int n)
{
Code
C
Auto
int smallestEvenMultiple(int n)
{
int result;
Saved
Ln 17, Col 1
Testcase
Test Result
Accepted Runtime: 0 ms
Case 1
Case 2
Input
n =
5
Output
10
Expected
10
Contribute a testcase

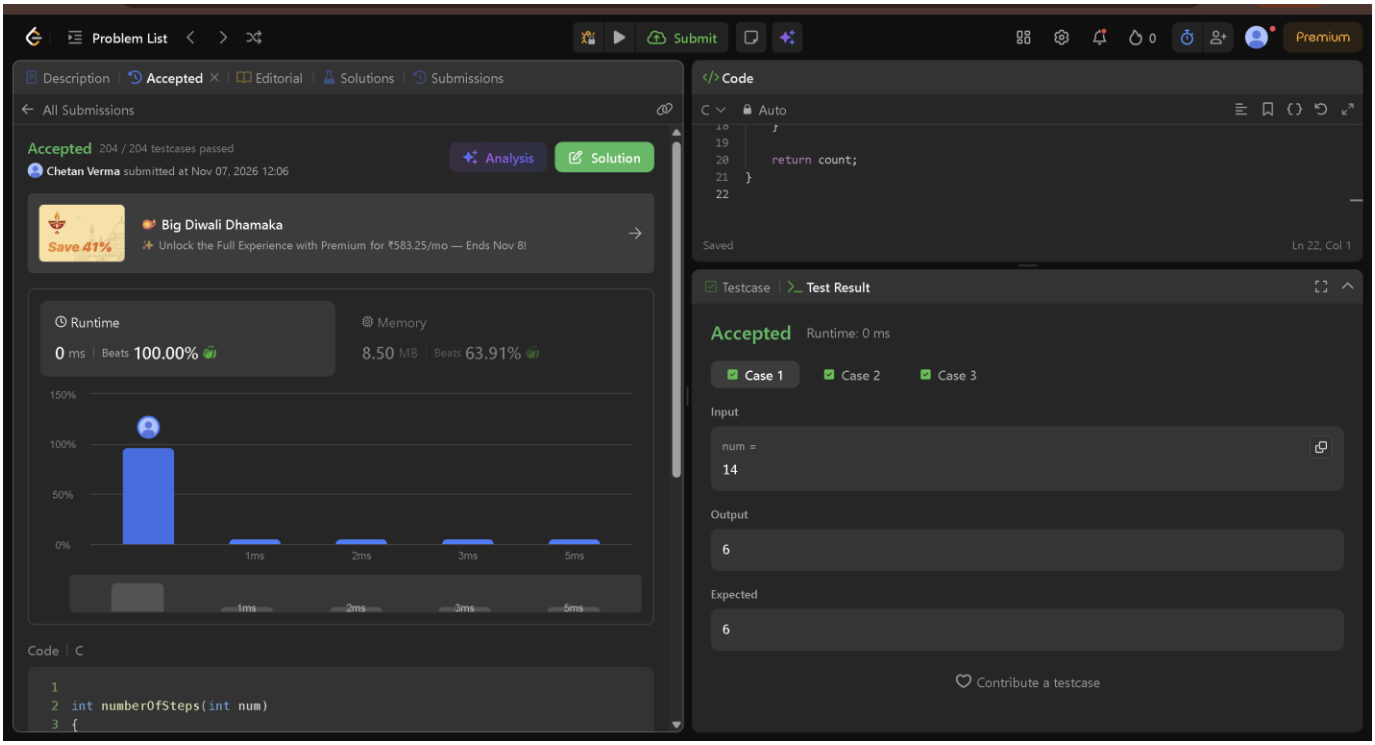

HackerRank
Prepare
Certify
Compete
Grading Students
Your Grading Students submission got 10.00 points.
You are now 19 points away from the 1st star for your problem solving badge.
Try the next challenge | Try a Random Challenge
Problem
Submissions
Leaderboard
Discussions
Editorial
You made this submission a few seconds ago.
Score: 10.00 Status: Accepted
People who solved Grading Students attempted this next:
Apple and Orange
Find the respective numbers of apples and oranges that fall on Sam's house.
Solve Challenge
Submitted Code
Language: C
Open in editor
NEED HELP?
View discussions
View editorial
View top submissions
Test case 0
Test case 1
Test case 2
Test case 3
Test case 4
Test case 5
Test case 6
Compiler Message
Success
Input (stdin)
Download
4
73
67
38
33
Expected Output
Download
75
67

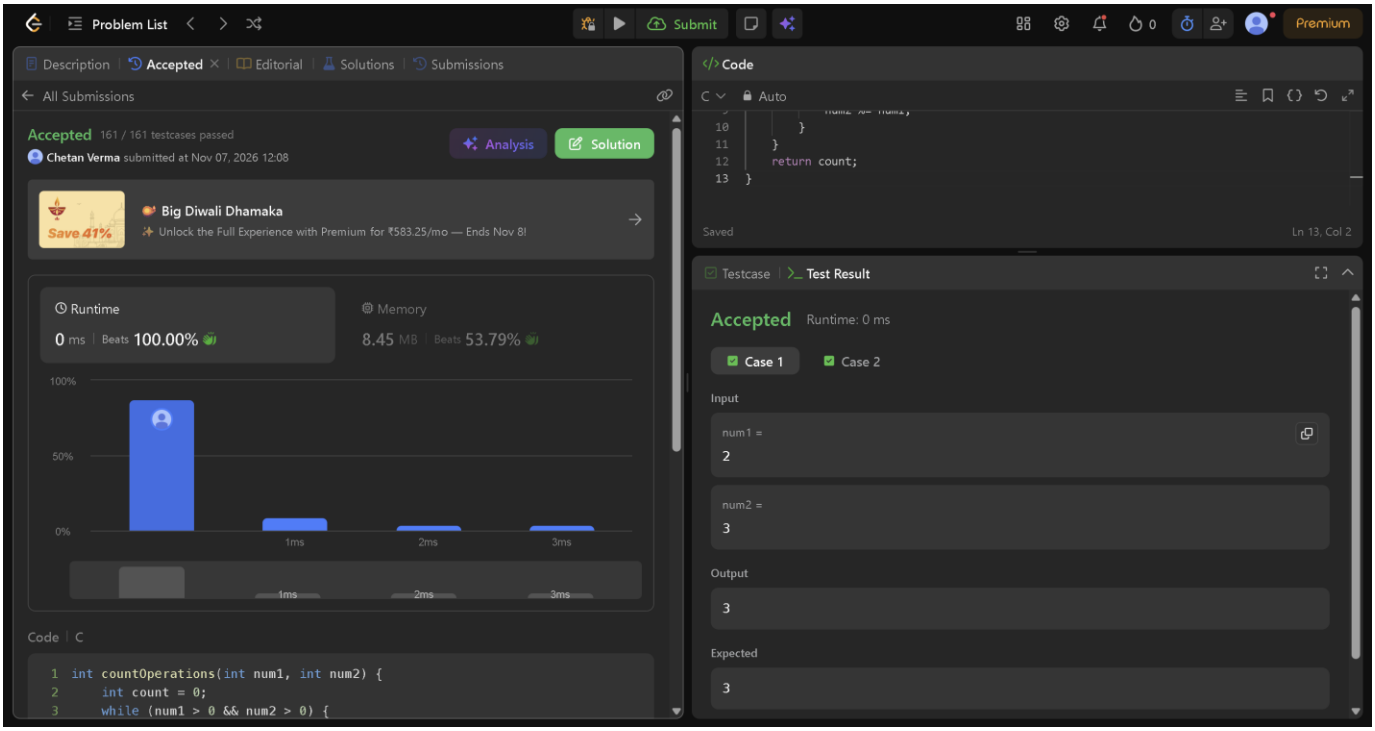

Submit
00:00:00
Description
Editorial
Submissions
Similar Problems
Discussion
C
Roy and Profile Picture
4.0 (1867 votes)
Ad Hoc
Approved
Basic programming
Easy
Implementation
Open
82.0% success
314,894 attempts
1s time
256MB memory
1024KB max code
Roy wants to change his profile picture on Facebook. Now Facebook has some restriction over the dimension of picture that we can upload.
Minimum dimension of the picture can be L x L, where L is the length of the side of square.
Now Roy has N photos of various dimensions.
Dimension of a photo is denoted as W x H
where W - width of the photo and H - Height of the photo
When any photo is uploaded following events may occur:
[1] If any of the width or height is less than L, user is prompted to upload another one. Print "UPLOAD ANOTHER" in this case.
[2] If width and height, both are large enough and
(a) if the photo is already square then it is accepted. Print "ACCEPTED" in this case.
(b) else user is prompted to crop it. Print "CROP IT" in this case.
(quotes are only for clarification)
Given L, N, W and H as input, print appropriate text as output.
Input:
First line contains L.
Second line contains N, number of photos.
Following N lines each contains two space separated integers W and H.
Output:
Print appropriate text for each photo in a new line.
Constraints:
1 <= L,W,H <= 10000
1 <= N <= 1000
Examples
#include <stdio.h>
int main()
{
int L, N;
int W, H;
scanf("%d", &L);
scanf("%d", &N);
for (int i = 0; i < N; i++)
{
scanf("%d %d", &W, &H);
if (W < L || H < L)
{
printf("UPLOAD ANOTHER\n");
}
Results
Test against custom input
Score
20
Time
0.066s
Memory
2KiB
Language
C
Input
Result
Time(sec)
Memory(KiB)
Score
Output
Expected
Diff
Test #1
Accepted
0.010
2
10
Test #2
Accepted
0.009
2
10
Test #3
Accepted
0.009
2
10
Test #4
Accepted
0.010
2
20
Test #5
Accepted
0.010
2
20
Test #6
Accepted
0.009
2
20
Test #7
Accepted
0.010
2
10

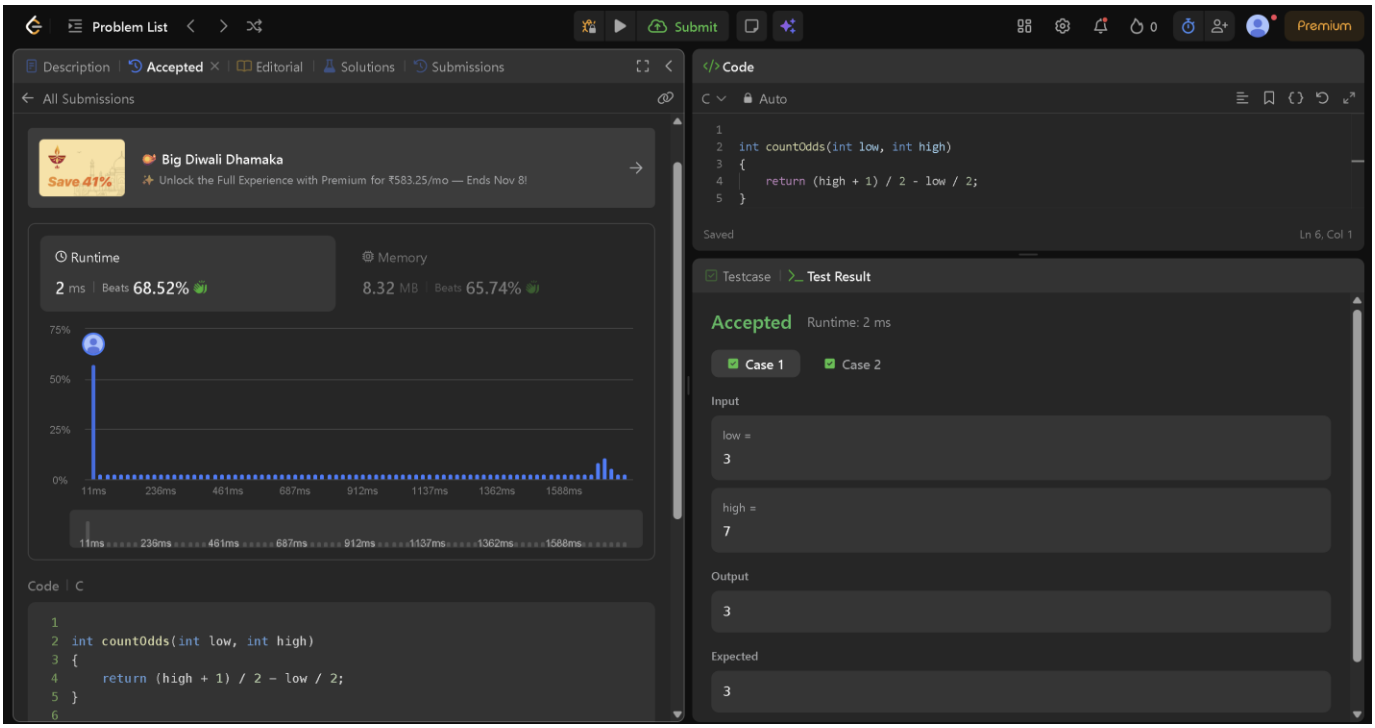

FizzBuzz
FizzBuzz
Problem
Submissions
Leaderboard
Discussions
You made this submission a few seconds ago.
Score: 2.00
Status: Accepted
Submitted Code
Language: C
Open in editor
NEED HELP?
View discussions
View top submissions
Test case 0
Compiler Message
0.1Success

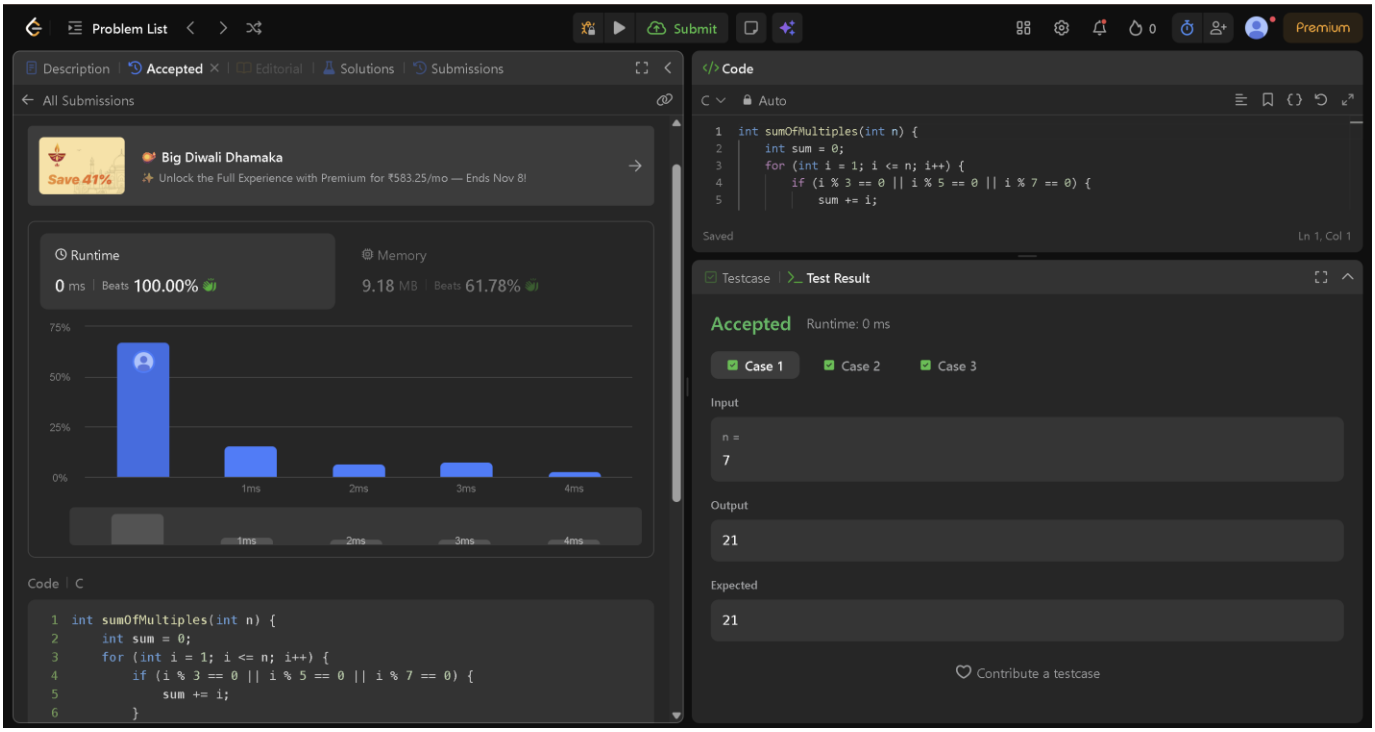
Module 3 — Loops and repeated computation

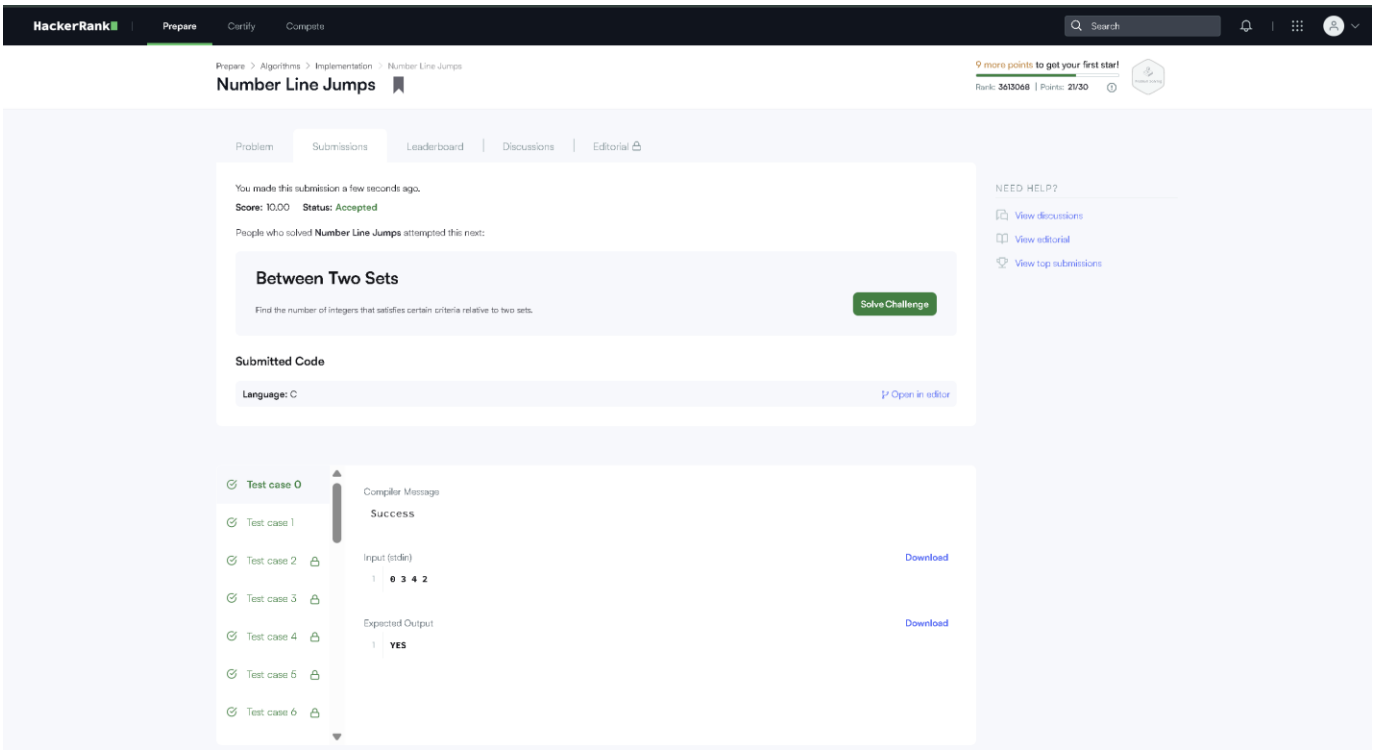




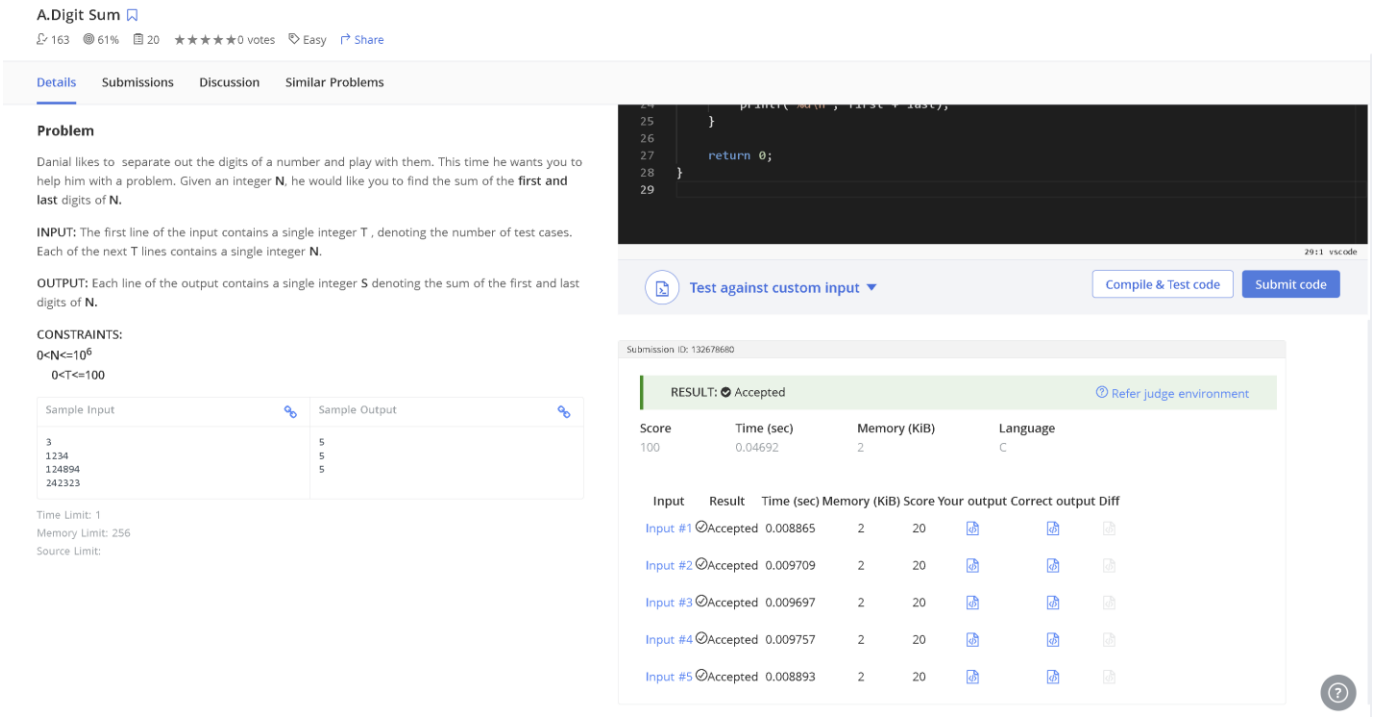

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Accepted 161 / 161 testcases passed
Chetan Verma submitted at Nov 07, 2026 12:08
Analysis
Solution
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
0 ms | Beats 100.00%
Memory
8.45 MB | Beats 53.79%
Code | C
int countOperations(int num1, int num2) {
int count = 0;
while (num1 > 0 && num2 > 0) {
Code
Auto
}
}
return count;
}
Saved
Ln 13, Col 2
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
num1 =
2
num2 =
3
Output
3
Expected
3

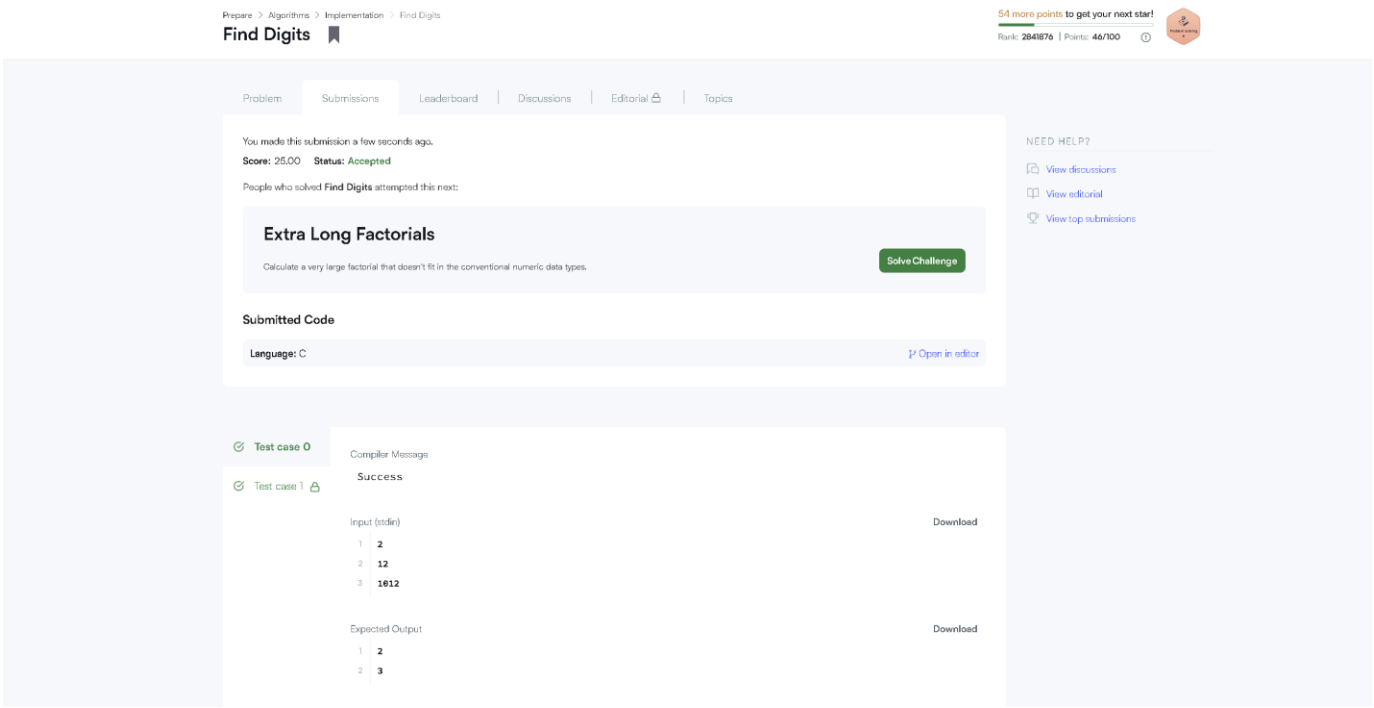

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
2 ms | Beats 68.52%
Memory
8.32 MB | Beats 65.74%
Code | C
int countOdds(int low, int high)
{
return (high + 1) / 2 - low / 2;
}
Code
Auto
Saved
Ln 6, Col 1
Testcase
Test Result
Accepted
Runtime: 2 ms
Case 1
Case 2
Input
low =
3
high =
7
Output
3
Expected
3

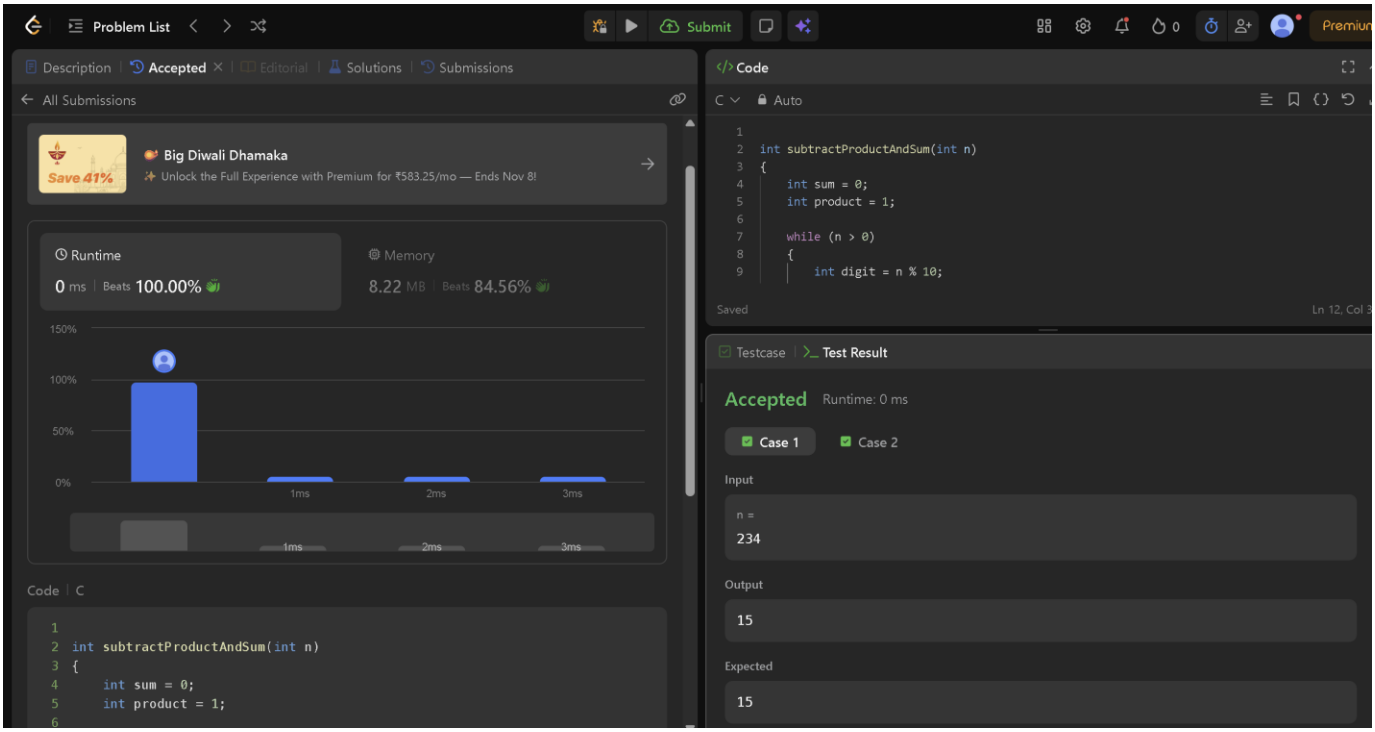

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
0 ms | Beats 100.00%
Memory
9.18 MB | Beats 61.78%
75%
50%
25%
0%
1ms
2ms
3ms
4ms
Code | C
int sumOfMultiples(int n) {
int sum = 0;
for (int i = 1; i <= n; i++) {
if (i % 3 == 0 || i % 5 == 0 || i % 7 == 0) {
sum += i;
}
Code
C
Auto
int sumOfMultiples(int n) {
int sum = 0;
for (int i = 1; i <= n; i++) {
if (i % 3 == 0 || i % 5 == 0 || i % 7 == 0) {
sum += i;
Saved
Ln 1, Col 1
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Input
n =
7
Output
21
Expected
21
Contribute a testcase

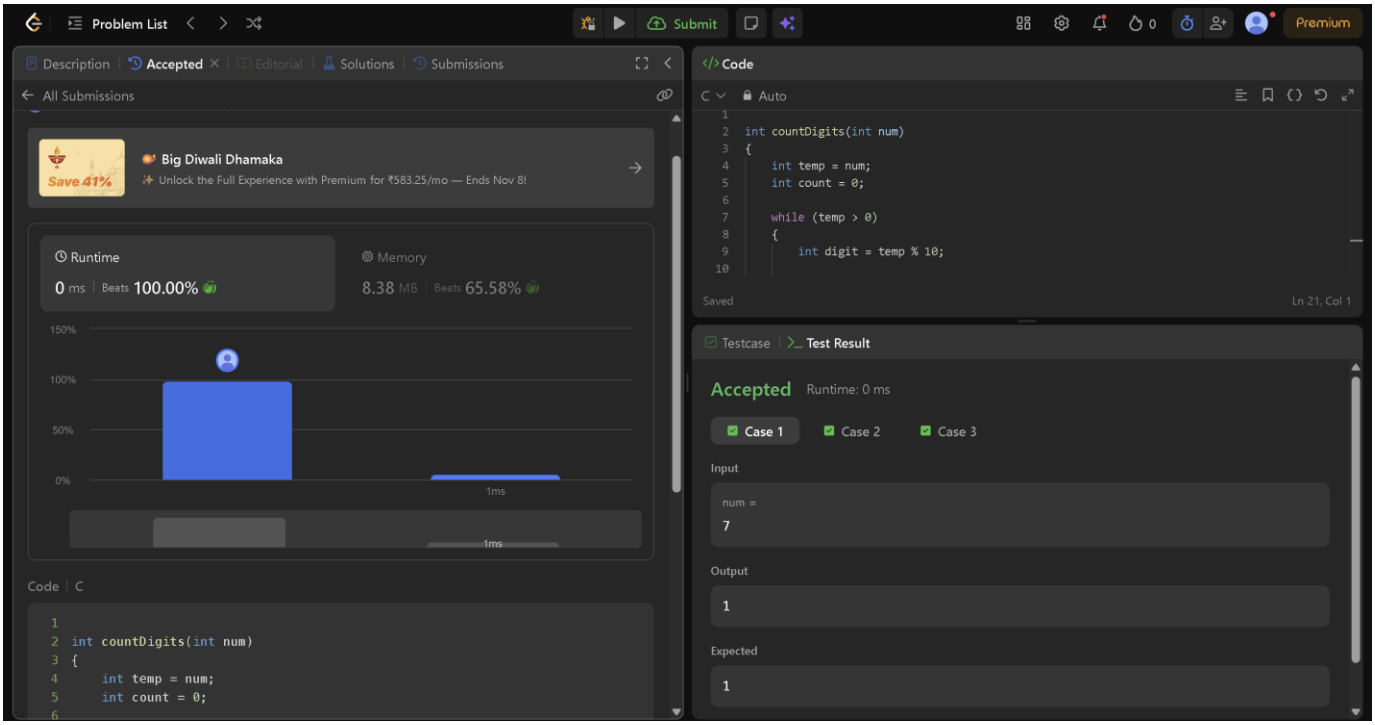

HackerRank
Prepare
Certify
Compete
Search
Prepare > Algorithms > Implementation > Number Line Jumps
Number Line Jumps
9 more points to get your first star!
Problem
Submissions
Leaderboard
Discussions
Editorial
You made this submission a few seconds ago.
Score: 10.00 Status: Accepted
People who solved Number Line Jumps attempted this next:
Between Two Sets
Find the number of integers that satisfies certain criteria relative to two sets.
Solve Challenge
Submitted Code
Language: C
Open in editor
NEED HELP?
View discussions
View editorial
View top submissions
Test case 0
Test case 1
Test case 2
Test case 3
Test case 4
Test case 5
Test case 6
Compiler Message
Success
Input (stdin)
Download
0 3 4 2
Expected Output
Download
YES

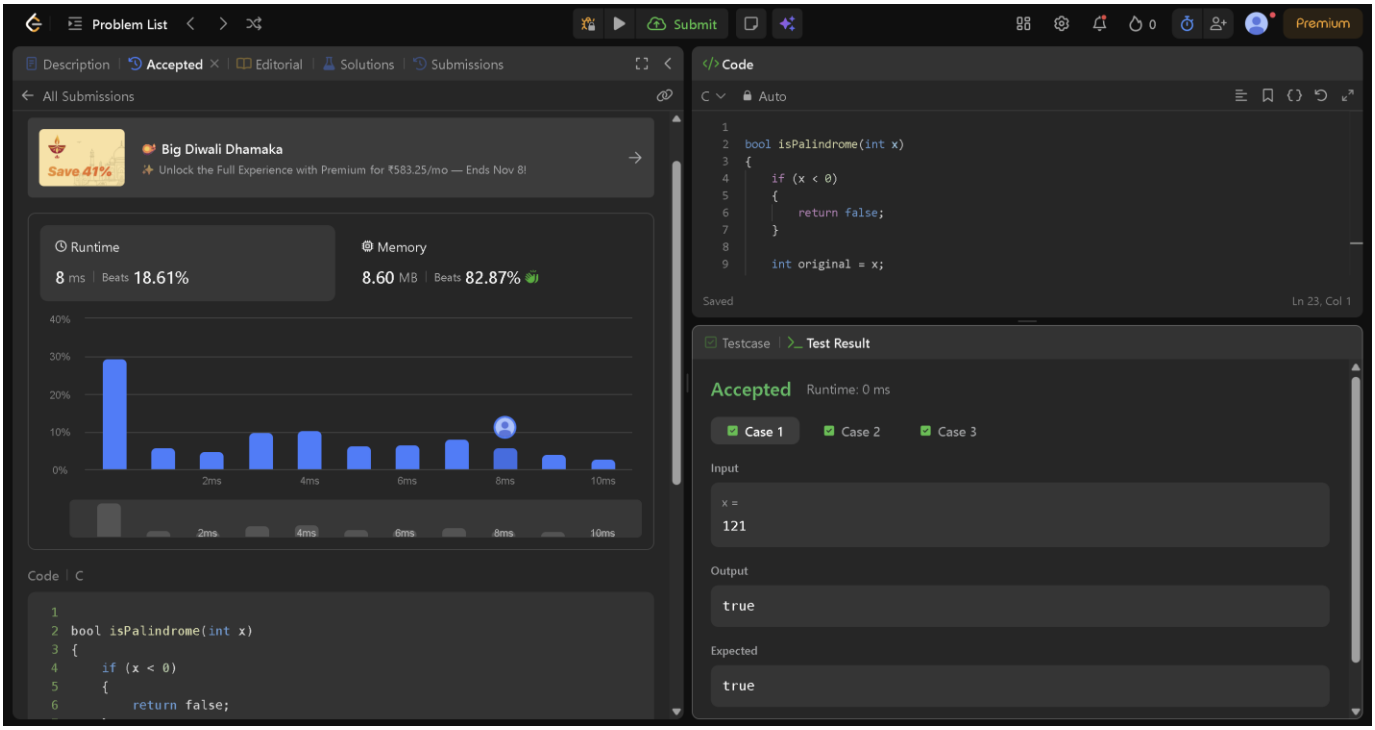
Module 4 — Digit extraction and number manipulation

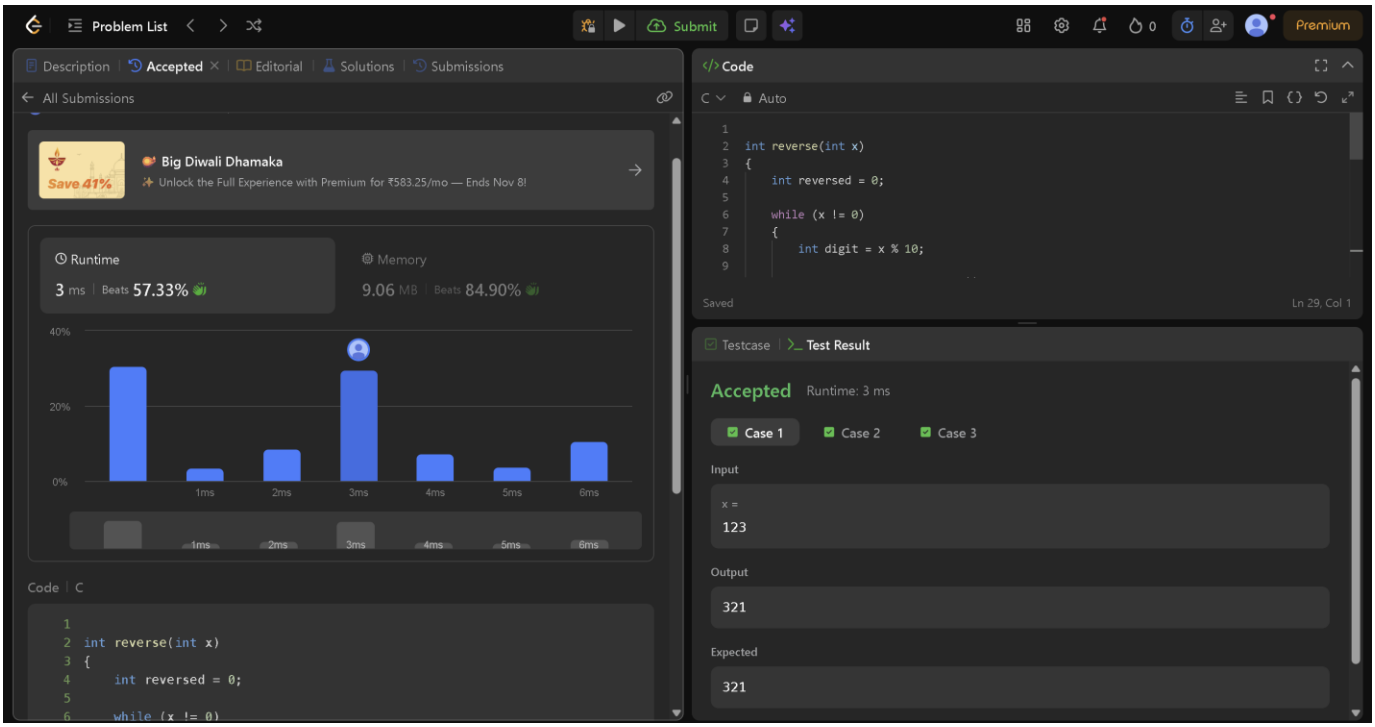




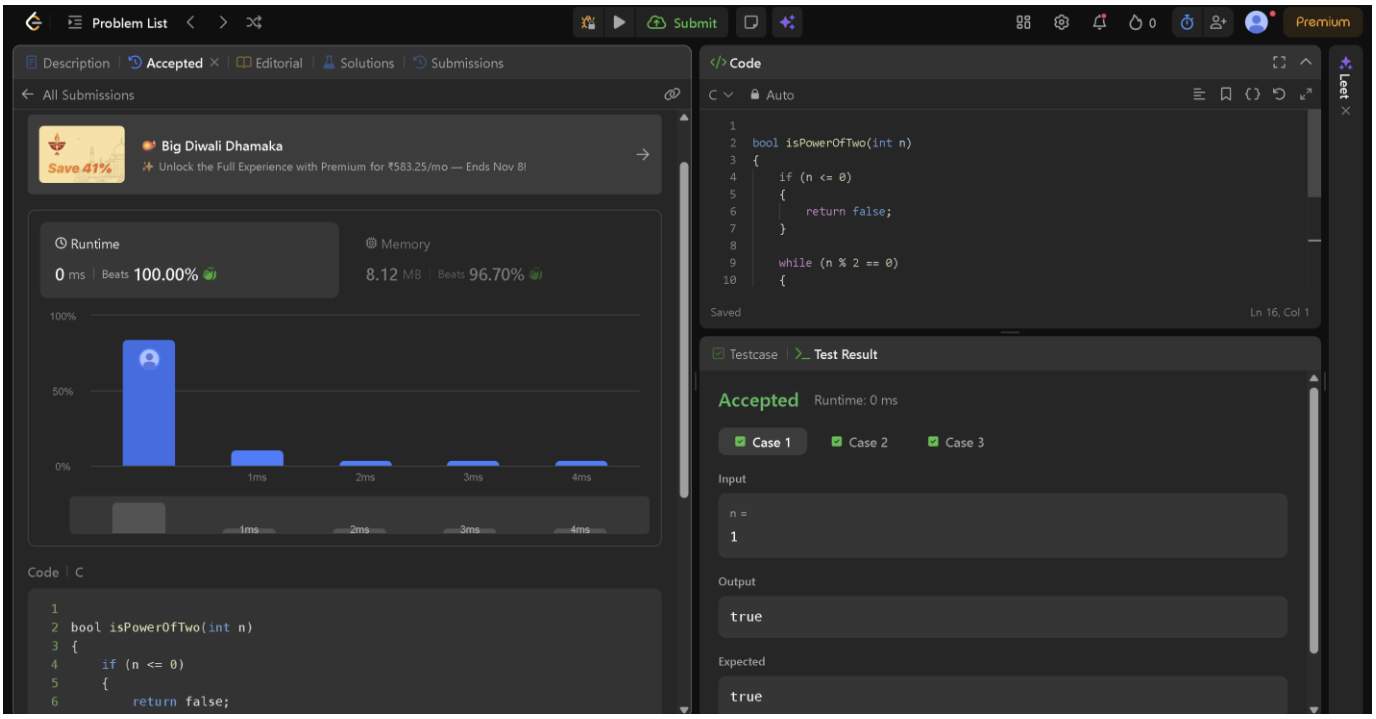

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Save 41%
Runtime
0 ms | Beats 100.00%
Memory
8.22 MB | Beats 84.56%
150%
100%
50%
0%
1ms
2ms
3ms
Code | C
int subtractProductAndSum(int n)
{
int sum = 0;
int product = 1;
Code
C
Auto
int subtractProductAndSum(int n)
{
int sum = 0;
int product = 1;
while (n > 0)
{
int digit = n % 10;
Saved
Ln 12, Col 3
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
n =
234
Output
15
Expected
15

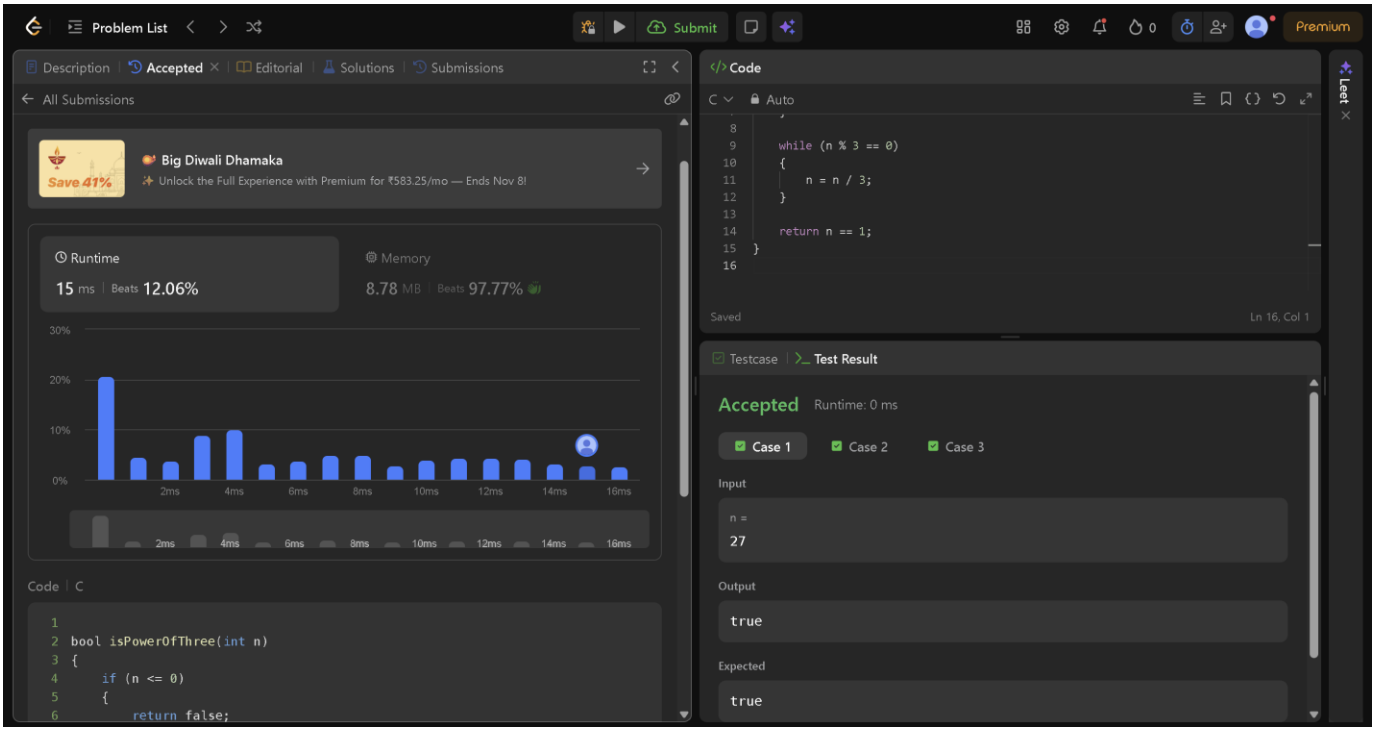

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Save 41%
Runtime
0 ms | Beats 100.00%
Memory
8.38 MB | Beats 65.58%
150%
100%
50%
0%
1ms
Code | C
int countDigits(int num)
{
int temp = num;
int count = 0;
Code
C
Auto
int countDigits(int num)
{
int temp = num;
int count = 0;
while (temp > 0)
{
int digit = temp % 10;
Saved
Ln 21, Col 1
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Input
num =
7
Output
1
Expected
1

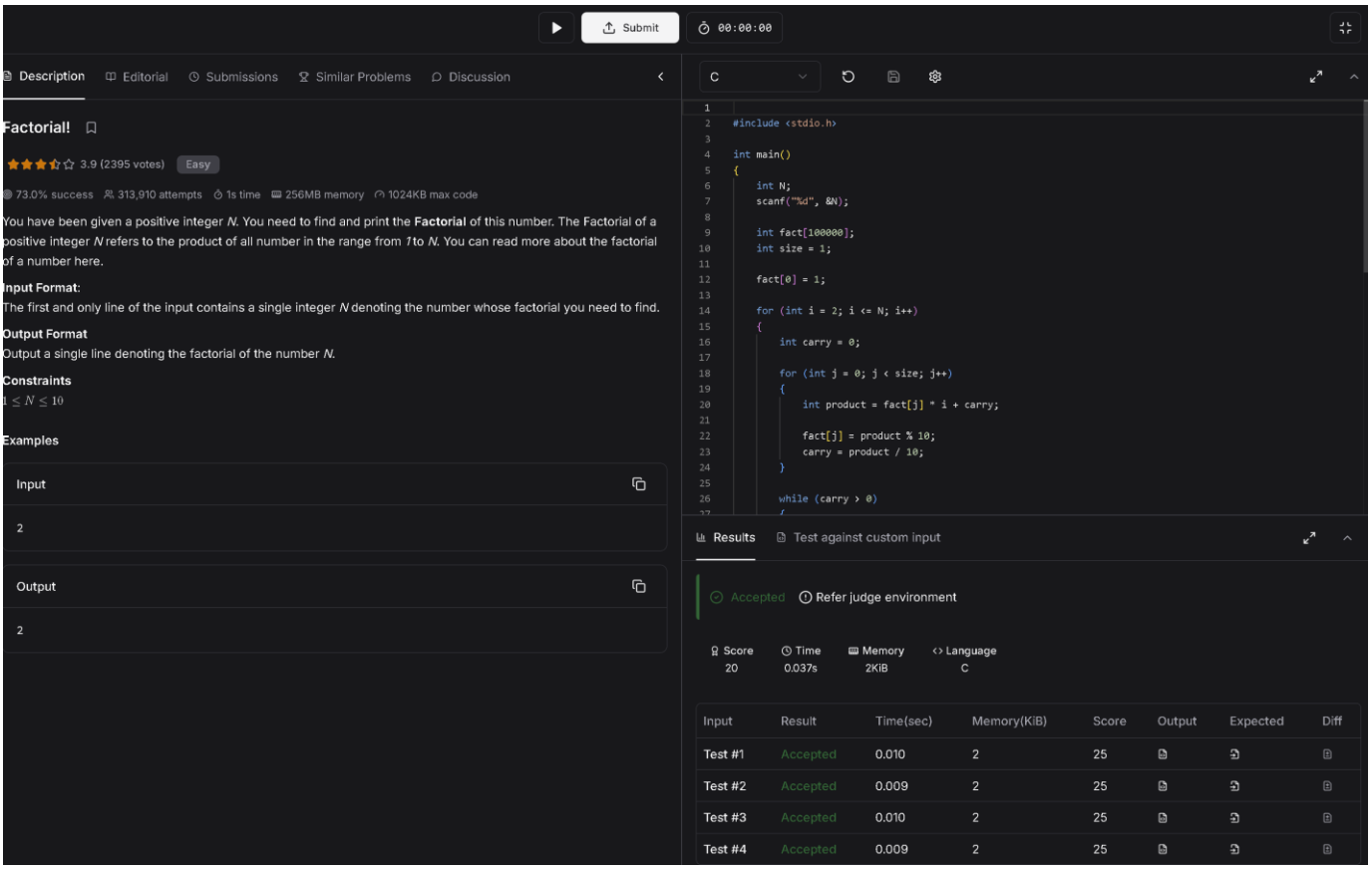

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
8 ms | Beats 18.61%
Memory
8.60 MB | Beats 82.87%
Code | C
bool isPalindrome(int x)
{
if (x < 0)
{
return false;
Code
C
Auto
bool isPalindrome(int x)
{
if (x < 0)
{
return false;
}
int original = x;
Saved
Ln 23, Col 1
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Input
x =
121
Output
true
Expected
true

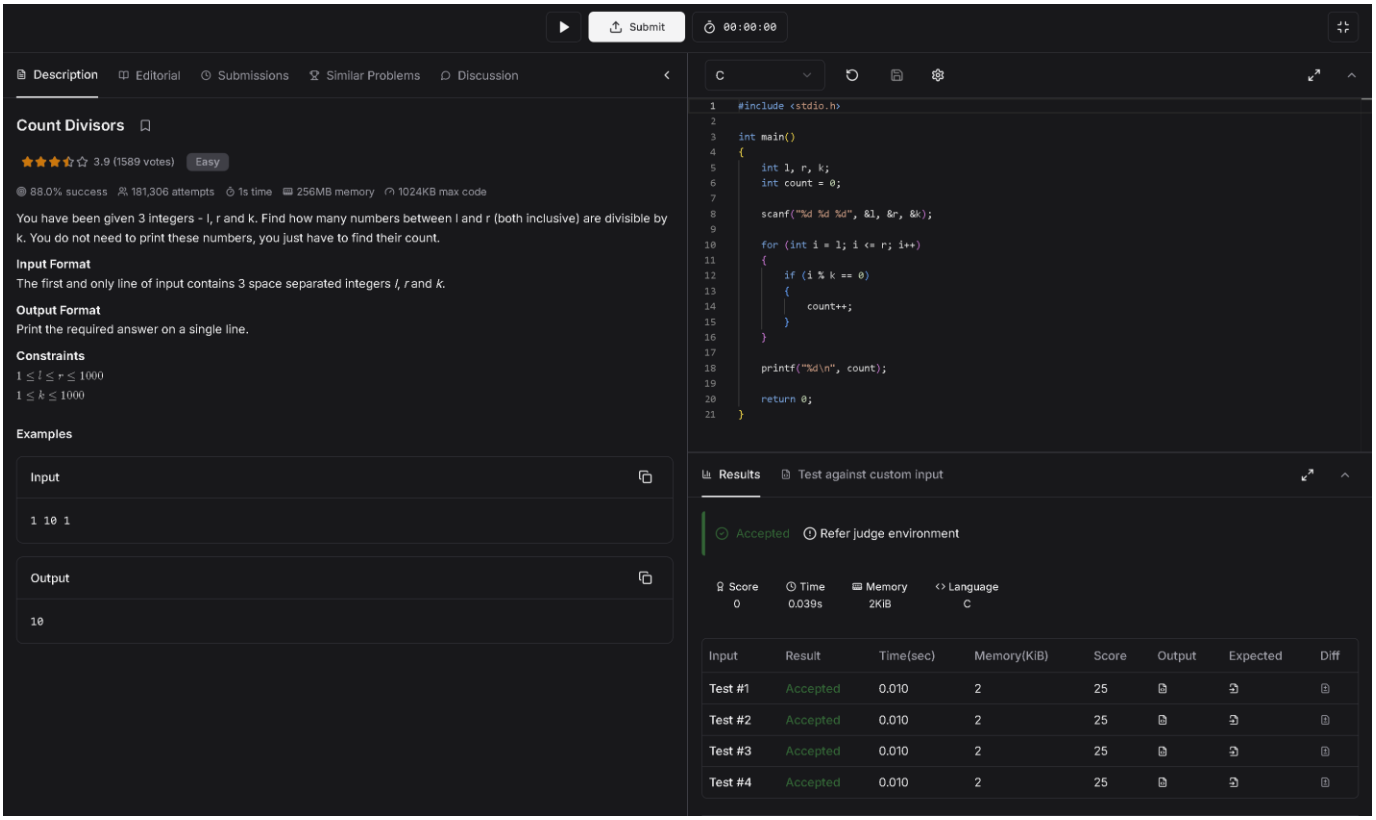

Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
3 ms | Beats 57.33%
Memory
9.06 MB | Beats 84.90%
Code | C
int reverse(int x)
{
int reversed = 0;
while (x
Code
C
Auto
int reverse(int x)
{
int reversed = 0;
while (x != 0)
{
int digit = x % 10;
Saved
Ln 29, Col 1
Testcase
Test Result
Accepted
Runtime: 3 ms
Case 1
Case 2
Case 3
Input
x =
123
Output
321
Expected
321

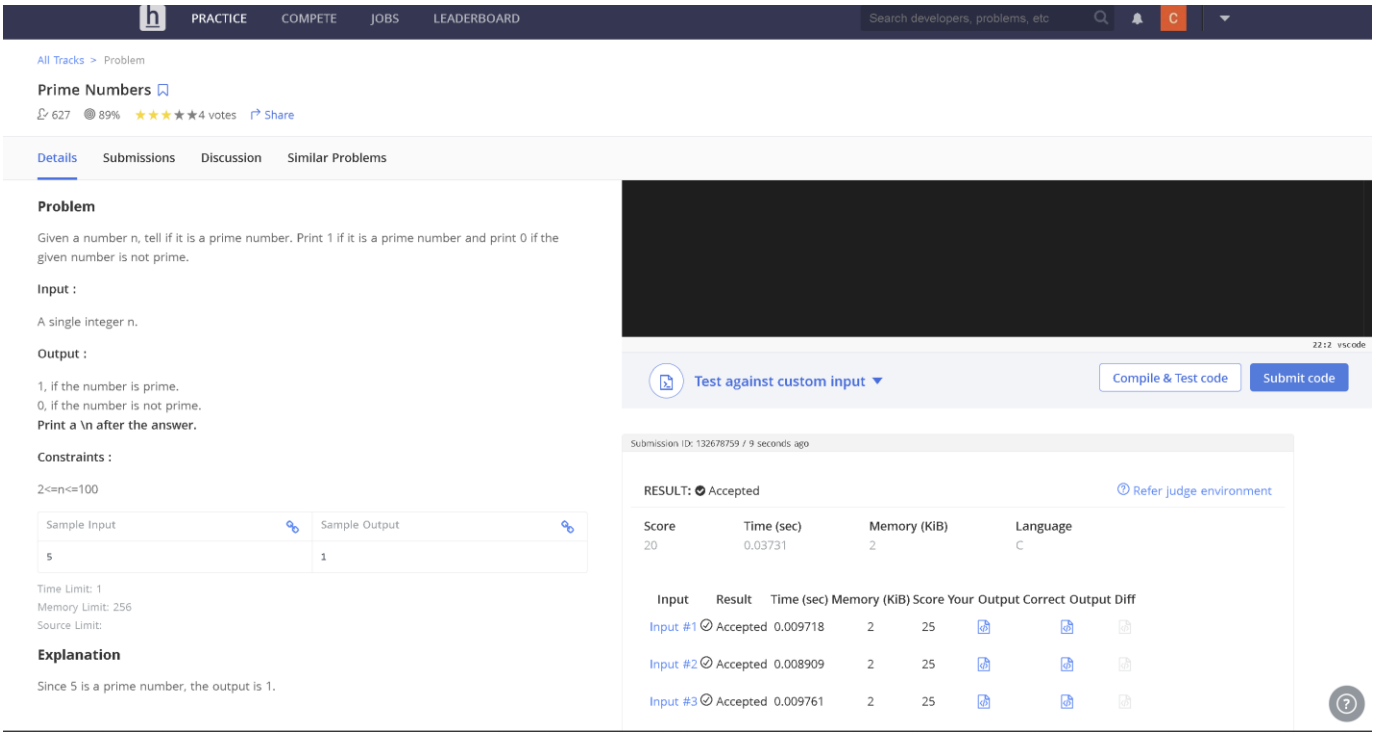
Module 5 — Factors, divisibility and mathematical loops

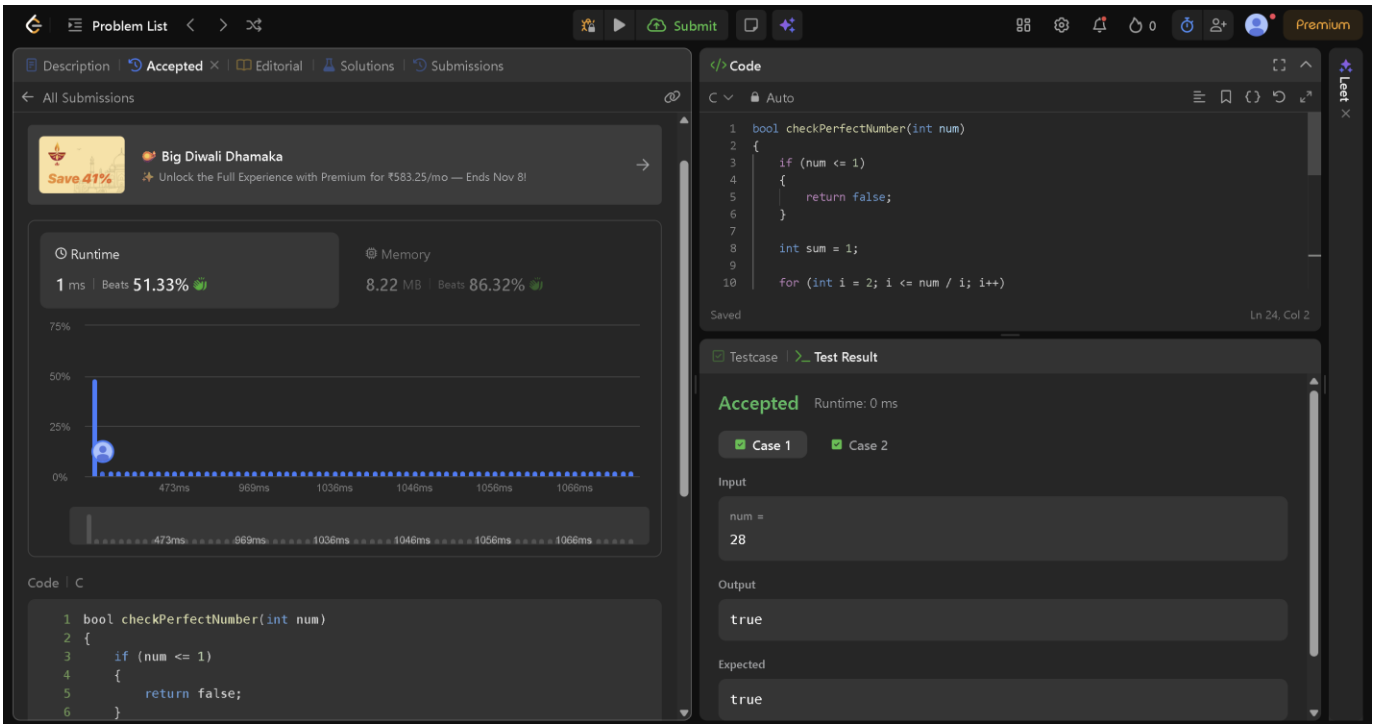




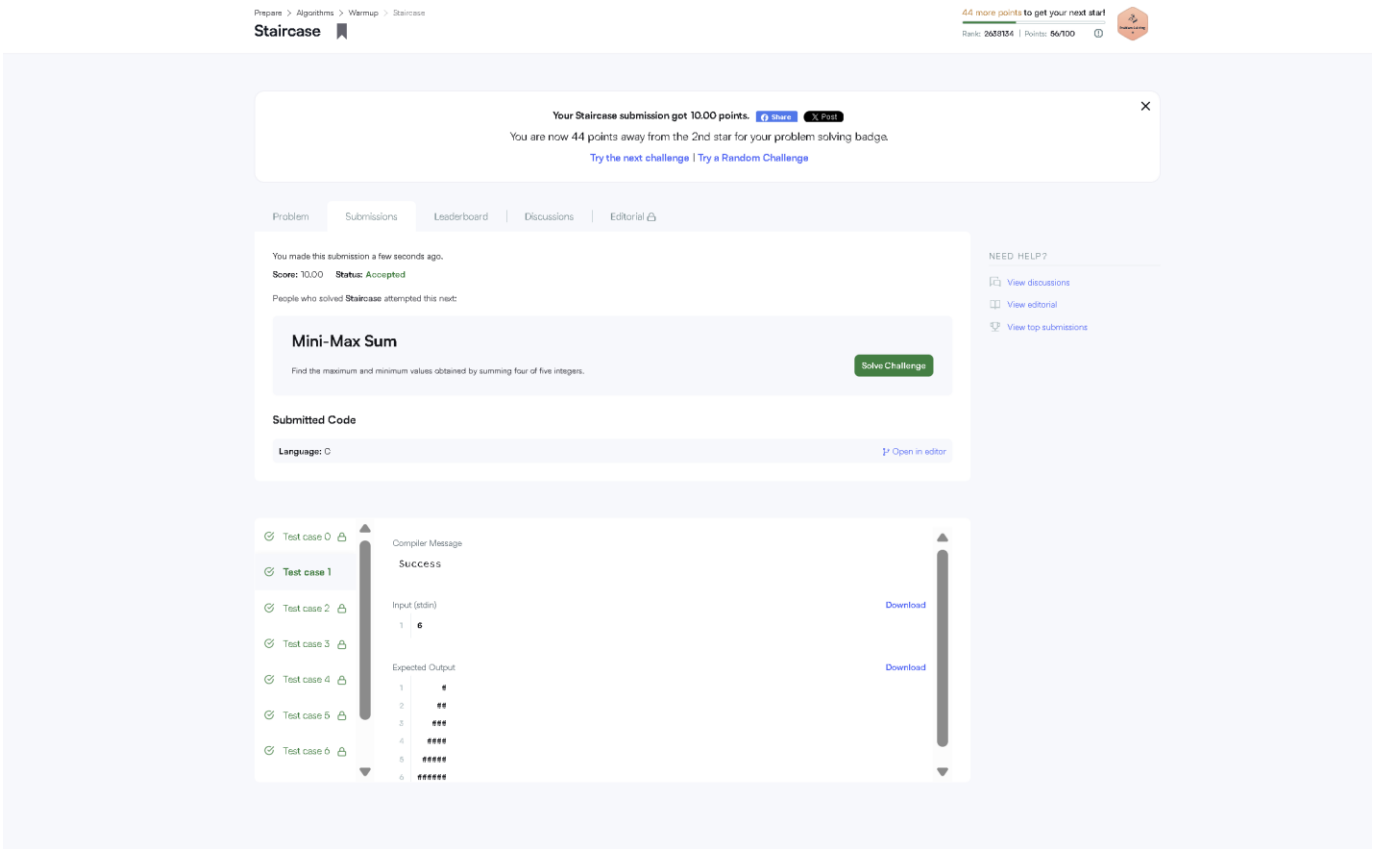

Submit
00:00:00
Description
Editorial
Submissions
Similar Problems
Discussion
C
Factorial!
3.9 (2395 votes)
Easy
73.0% success
313,910 attempts
1s time
256MB memory
1024KB max code
You have been given a positive integer N. You need to find and print the Factorial of this number. The Factorial of a positive integer N refers to the product of all number in the range from 1 to N. You can read more about the factorial of a number here.
Input Format:
The first and only line of the input contains a single integer N denoting the number whose factorial you need to find.
Output Format
Output a single line denoting the factorial of the number N.
Constraints
$1 \leq N \leq 10$
Examples
Input
2
Output
2
#include <stdio.h>
int main()
{
int N;
scanf("%d", &N);
int fact[100000];
int size = 1;
fact[0] = 1;
for (int i = 2; i <= N; i++)
{
int carry = 0;
for (int j = 0; j < size; j++)
{
int product = fact[j] * i + carry;
fact[j] = product % 10;
carry = product / 10;
}
while (carry > 0)
Results
Test against custom input
Accepted
Refer judge environment
Score
20
Time
0.037s
Memory
2KiB
Language
C
Input
Result
Time(sec)
Memory(KiB)
Score
Output
Expected
Diff
Test #1
Accepted
0.010
2
25
Test #2
Accepted
0.009
2
25
Test #3
Accepted
0.010
2
25
Test #4
Accepted
0.009
2
25

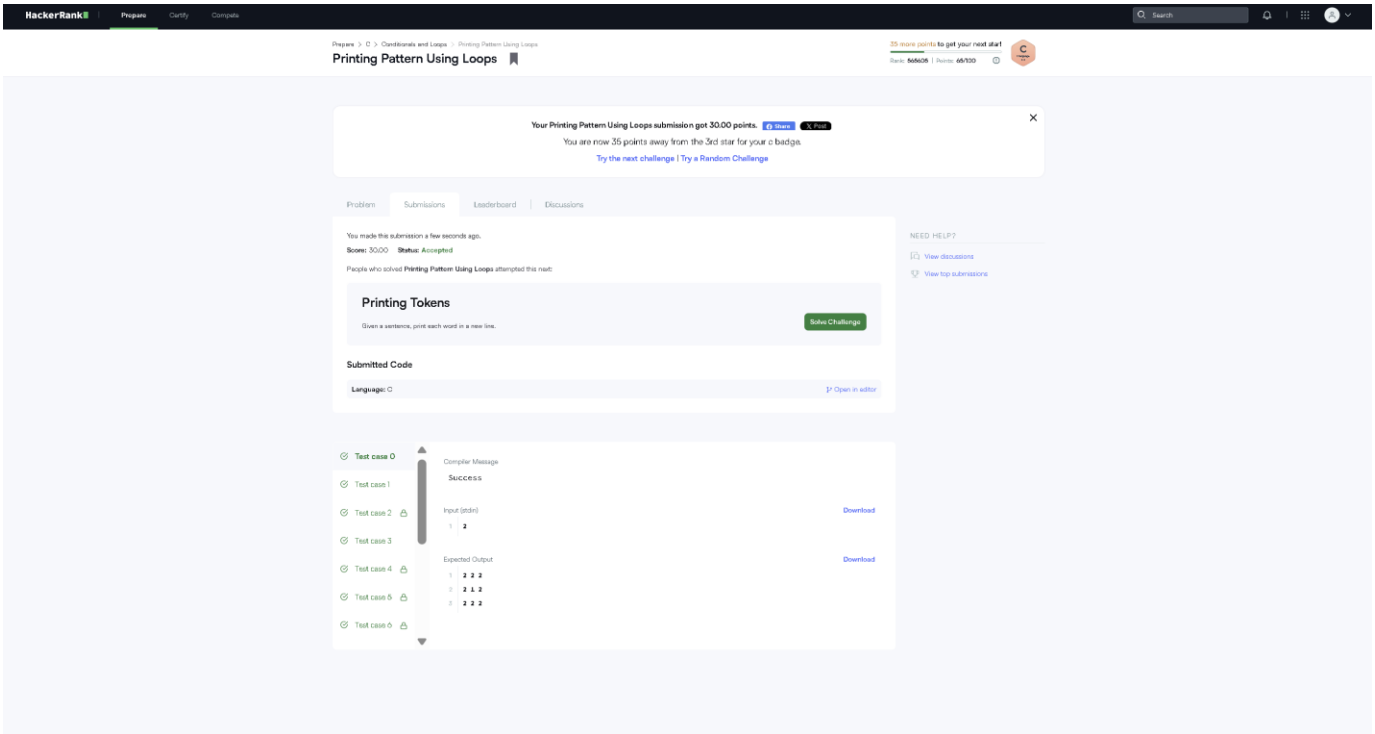

Submit
00:00:00
Description
Editorial
Submissions
Similar Problems
Discussion
C
Count Divisors
3.9 (1589 votes)
Easy
88.0% success
181,306 attempts
1s time
256MB memory
1024KB max code
You have been given 3 integers - l, r and k. Find how many numbers between l and r (both inclusive) are divisible by k. You do not need to print these numbers, you just have to find their count.
Input Format
The first and only line of input contains 3 space separated integers l, r and k.
Output Format
Print the required answer on a single line.
Constraints
$1 \leq l \leq r \leq 1000$
$1 \leq k \leq 1000$
Examples
Input
1 10 1
Output
10
#include <stdio.h>
int main()
{
int l, r, k;
int count = 0;
scanf("%d %d %d", &l, &r, &k);
for (int i = l; i <= r; i++)
{
if (i % k == 0)
{
count++;
}
}
printf("%d\n", count);
return 0;
}
Results
Test against custom input
Accepted
Refer judge environment
Score
0
Time
0.039s
Memory
2KiB
Language
C
Input
Result
Time(sec)
Memory(KiB)
Score
Output
Expected
Diff
Test #1
Accepted
0.010
2
25
Test #2
Accepted
0.010
2
25
Test #3
Accepted
0.010
2
25
Test #4
Accepted
0.010
2
25


PRACTICE
COMPETE
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Search developers, problems, etc
All Tracks > Problem
Prime Numbers
627
89%
4 votes
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Details
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Discussion
Similar Problems
Problem
Given a number n, tell if it is a prime number. Print 1 if it is a prime number and print 0 if the given number is not prime.
Input :
A single integer n.
Output :
1, if the number is prime.
0, if the number is not prime.
Print a \n after the answer.
Constraints :
2<=n<=100
Sample Input
Sample Output
5
1
Time Limit: 1
Memory Limit: 256
Source Limit:
Explanation
Since 5 is a prime number, the output is 1.
22:2 vscode
Test against custom input
Compile & Test code
Submit code
Submission ID: 132678759 / 9 seconds ago
RESULT: Accepted
Refer judge environment
Score
Time (sec)
Memory (KiB)
Language
20
0.03731
2
C
Input
Result
Time (sec)
Memory (KiB)
Score
Your Output
Correct Output
Diff
Input #1 Accepted 0.009718 2 25
Input #2 Accepted 0.008909 2 25
Input #3 Accepted 0.009761 2 25


Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
1 ms | Beats 51.33%
Memory
8.22 MB | Beats 86.32%
75%
50%
25%
0%
473ms
969ms
1036ms
1046ms
1056ms
1066ms
Code | C
bool checkPerfectNumber(int num)
{
if (num <= 1)
{
return false;
}
Code
Auto
bool checkPerfectNumber(int num)
{
if (num <= 1)
{
return false;
}
int sum = 1;
for (int i = 2; i <= num / i; i++)
Saved
Ln 24, Col 2
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Input
num =
28
Output
true
Expected
true

Module 6 — Nested loops: online judge problems





Module 7 — Nested-loop and pattern-printing questions

Question 1 — Print a Solid Square

Difficulty: Very Easy

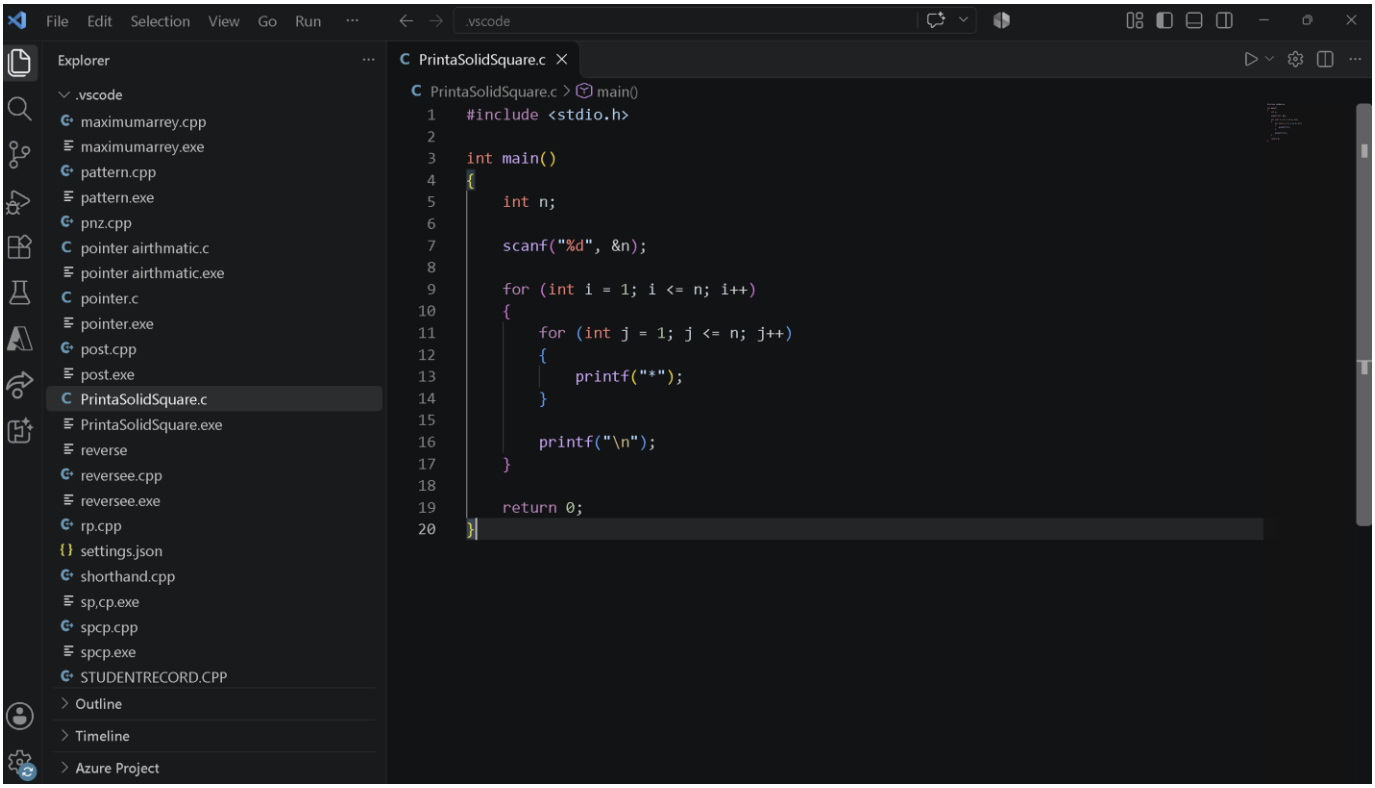
Problem statement: Given a positive integer n, print a square containing n rows and n columns of asterisks (*). Do not print spaces between asterisks.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, each containing exactly n asterisks.

Sample Input 4	**Sample Output** **** **** **** ****

Hint: Use one loop for rows and one loop for columns.



Question 2 — Print a Right-Angled Triangle

Difficulty: Very Easy

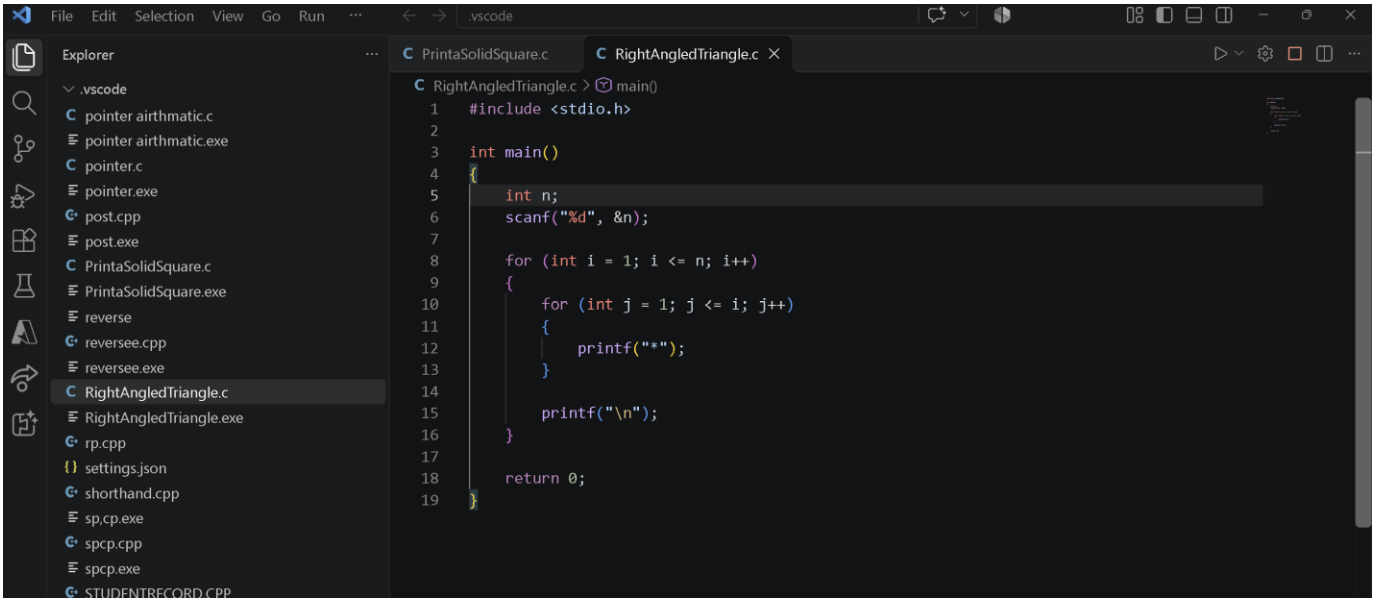
Problem statement: Given n, print a left-aligned triangle. Row i must contain exactly i asterisks, for i from 1 to n.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Line i contains i asterisks.

Sample Input 4	**Sample Output** * ** *** ****

Hint: The inner loop limit depends on the current row.



Question 3 — Print an Inverted Triangle

Difficulty: Easy

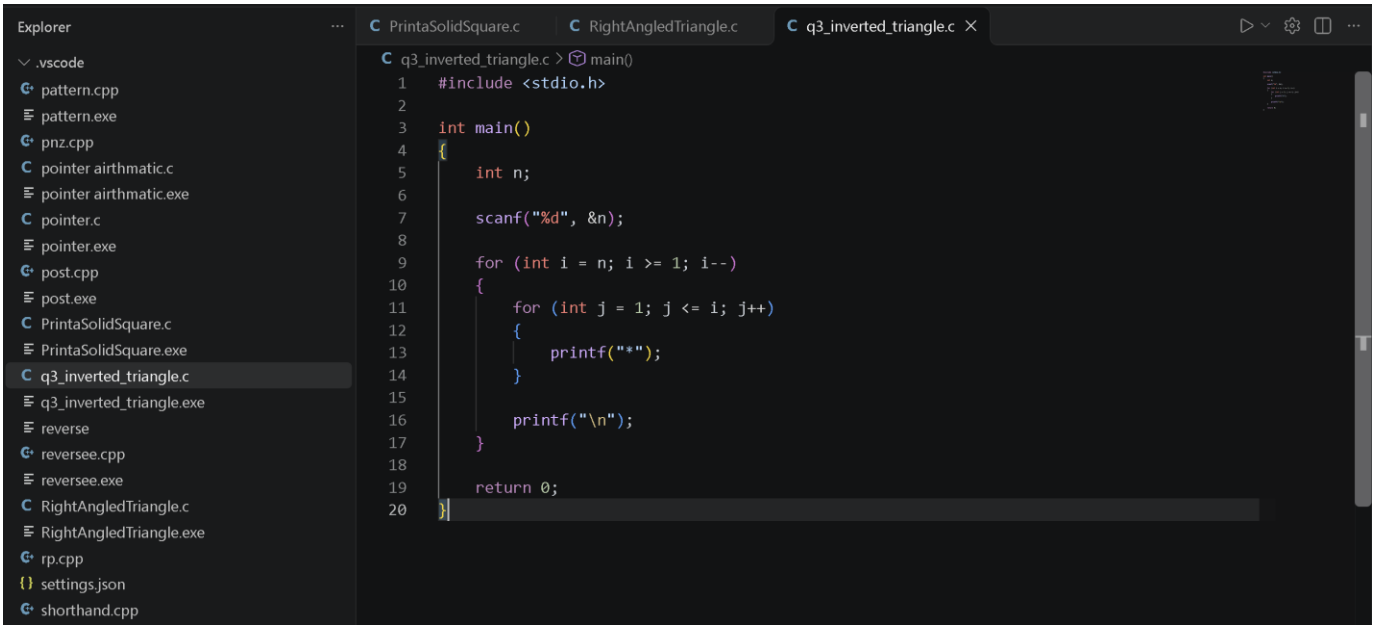
Problem statement: Given n, print a left-aligned inverted triangle. The first row contains n asterisks; each following row contains one fewer.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with the number of asterisks decreasing from n to 1.

Sample Input 4	**Sample Output** **** *** ** *

Hint: For row i, print n - i + 1 asterisks.



Question 4 — Print a Number Triangle

Difficulty: Easy

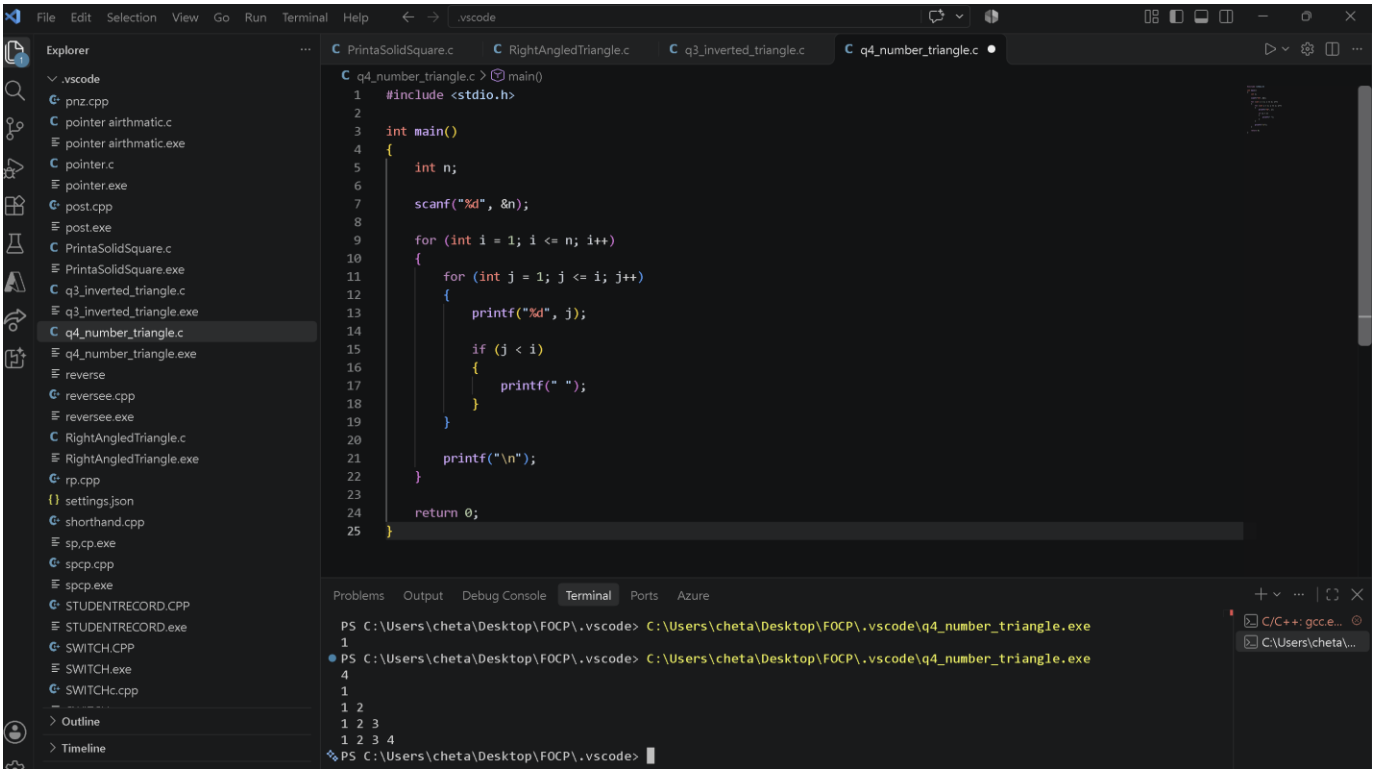
Problem statement: Given n, print n rows. On row i, print integers from 1 through i, separated by one space.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines with numbers separated by one space and no trailing spaces.

Sample Input 4	**Sample Output** 1 1 2 1 2 3 1 2 3 4

Hint: The inner loop prints from 1 through the current row number.



Question 5 — Print a Repeated-Number Triangle

Difficulty: Easy

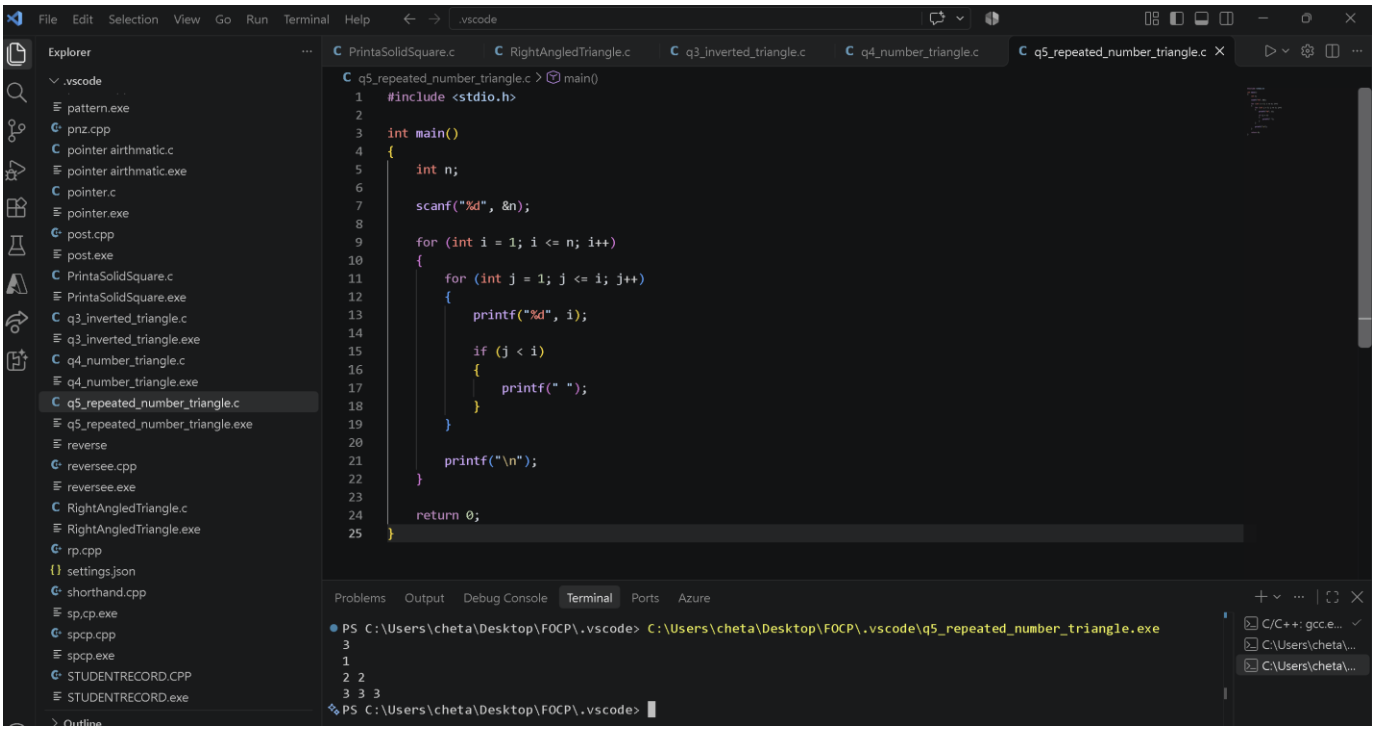
Problem statement: Given n, print n rows. On row i, print the number i exactly i times, separated by one space.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines; row i contains i copies of i.

Sample Input 4	**Sample Output** 1 2 2 3 3 3 4 4 4 4

Hint: Print the row counter during each inner-loop iteration.



Question 6 — Print a Consecutive-Number Triangle

Difficulty: Easy

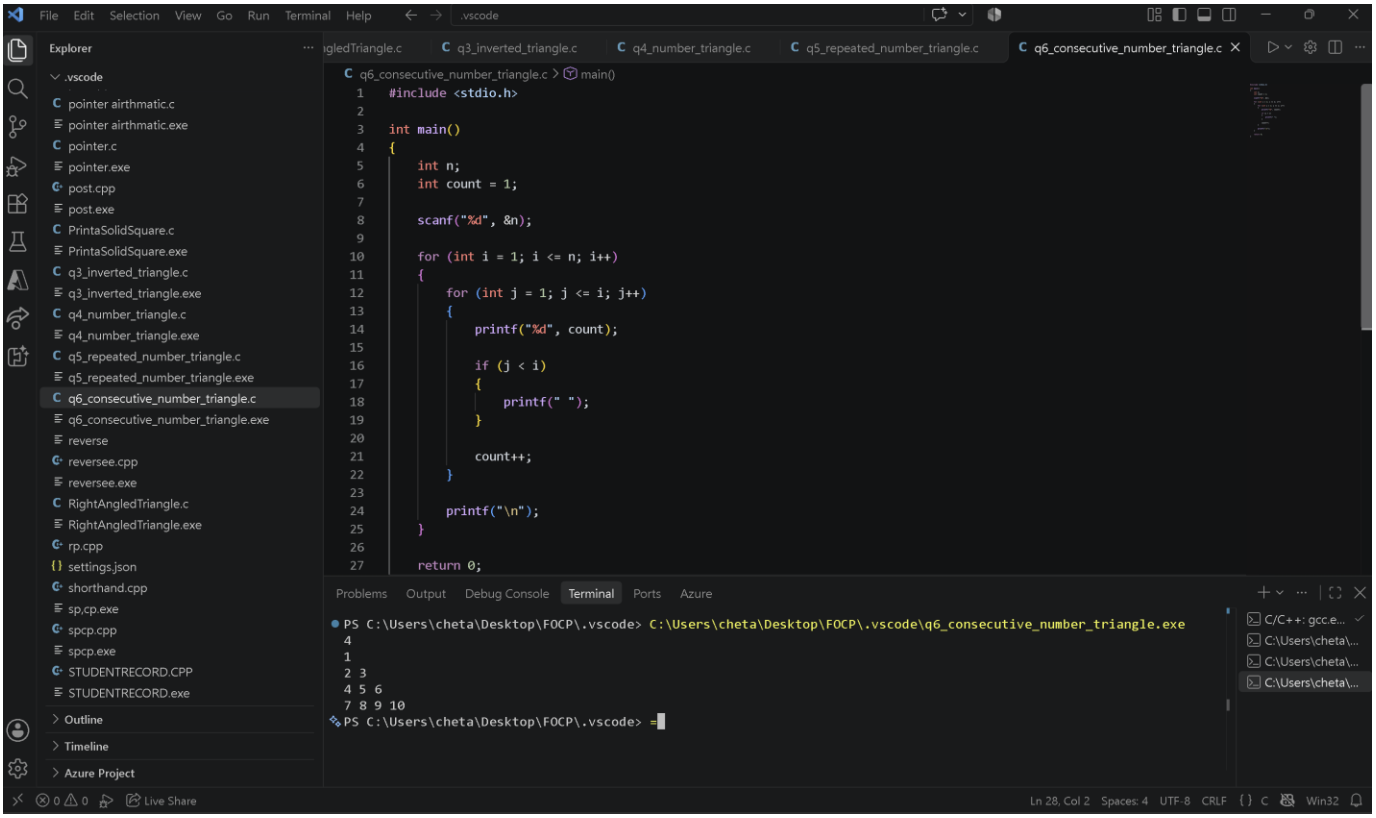
Problem statement: Print consecutive positive integers across triangular rows. Row 1 has one number, row 2 has two, and so on. Counting starts at 1 and continues across rows.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with values separated by one space.

Sample Input 4	**Sample Output** 1 2 3 4 5 6 7 8 9 10

Hint: Maintain one counter that increases after each printed number.



Question 7 — Print a Right-Aligned Star Triangle

Difficulty: Easy

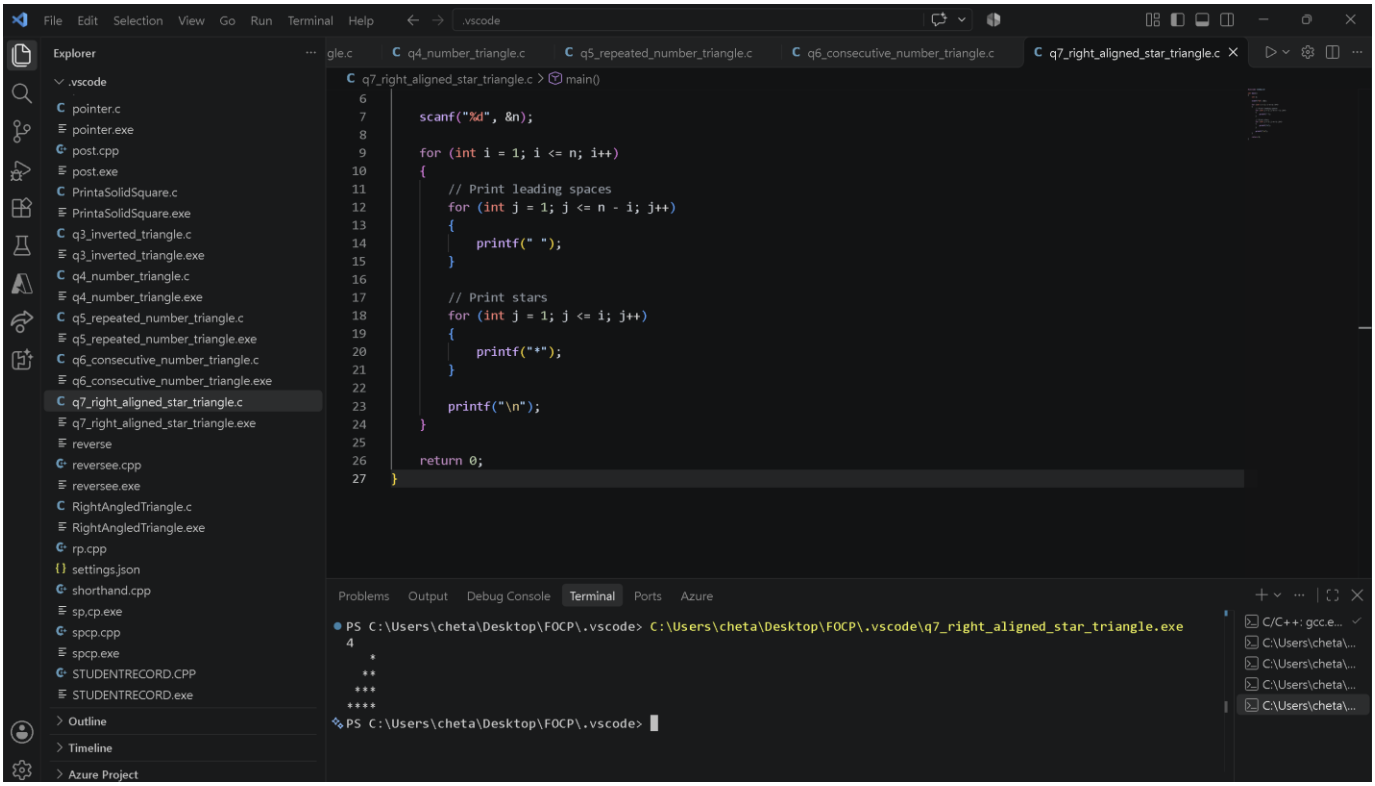
Problem statement: Given n, print a right-aligned triangle. Row 1 has one asterisk and row n has n asterisks. Use leading spaces to align the right edges.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Leading spaces are required; do not add trailing spaces.

Sample Input 4	**Sample Output** * ** *** ****

Hint: Row i has n - i leading spaces and i asterisks.



Question 8 — Print a Centered Pyramid

Difficulty: Easy–Moderate

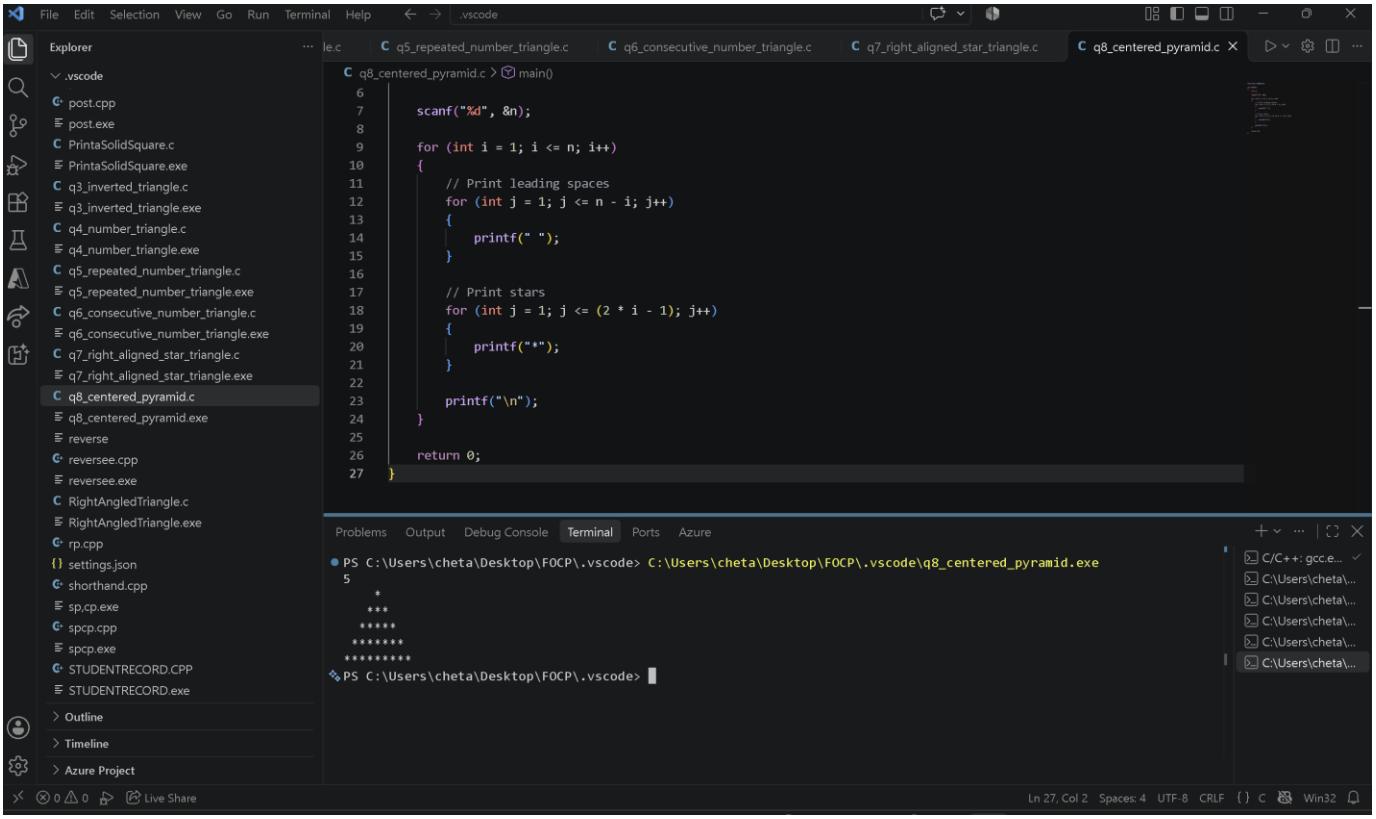
Problem statement: Given n, print a centered pyramid with n rows. Row i contains 2i - 1 asterisks, preceded by enough spaces to center it.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of leading spaces followed by an odd number of asterisks.

Sample Input 4	**Sample Output** * *** ***** *******

Hint: For row i, print n - i spaces and then 2*i - 1 asterisks.



Question 9 — Print an Inverted Centered Pyramid

Difficulty: Easy–Moderate

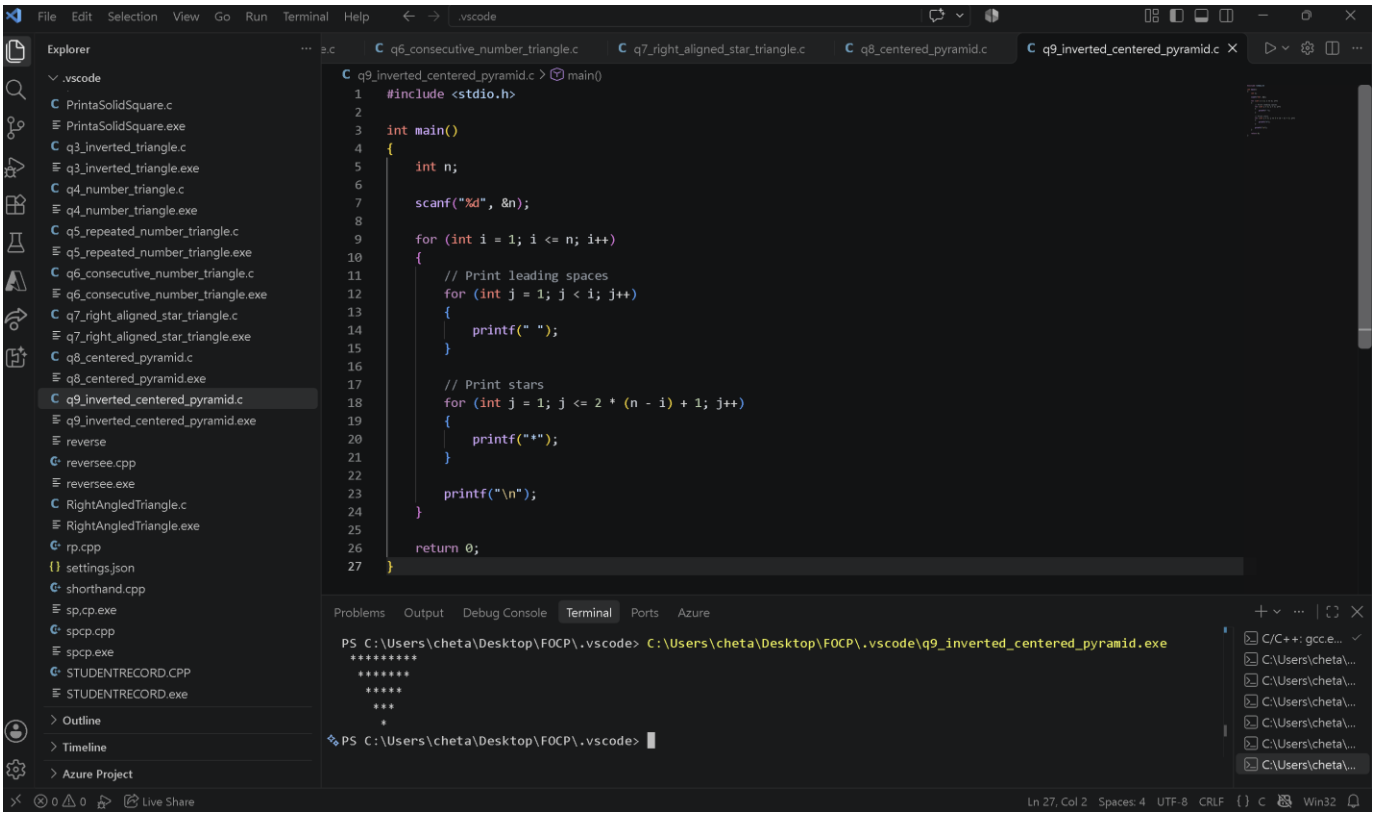
Problem statement: Given n, print an inverted centered pyramid. The first row contains 2n - 1 asterisks; each next row has two fewer and one more leading space.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with the pyramid narrowing toward the bottom.

Sample Input 4	**Sample Output** ******* ***** *** *

Hint: For row i, print i - 1 spaces and 2*(n - i) + 1 asterisks.



Question 10 — Print a Hollow Square

Difficulty: Moderate

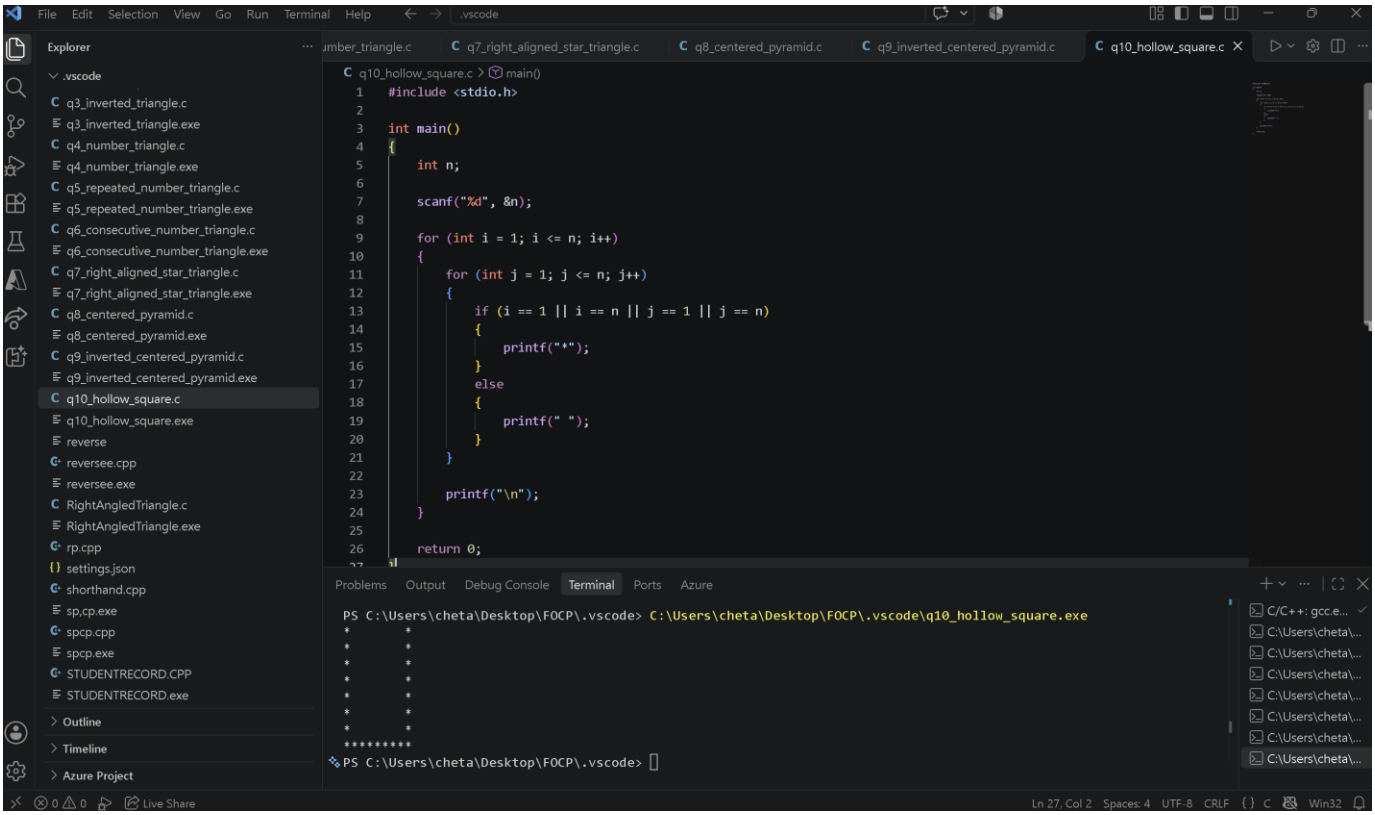
Problem statement: Given n, print an n-by-n hollow square. Print an asterisk on the first row, last row, first column, or last column. Print a space inside.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of width n; interior positions are spaces.

Sample Input 5	**Sample Output** ***** * * * * * * *****

Hint: Print '*' when the row or column is on a boundary; otherwise print a space.



Question 11 — Print a Hollow Right-Angled Triangle

Difficulty: Moderate

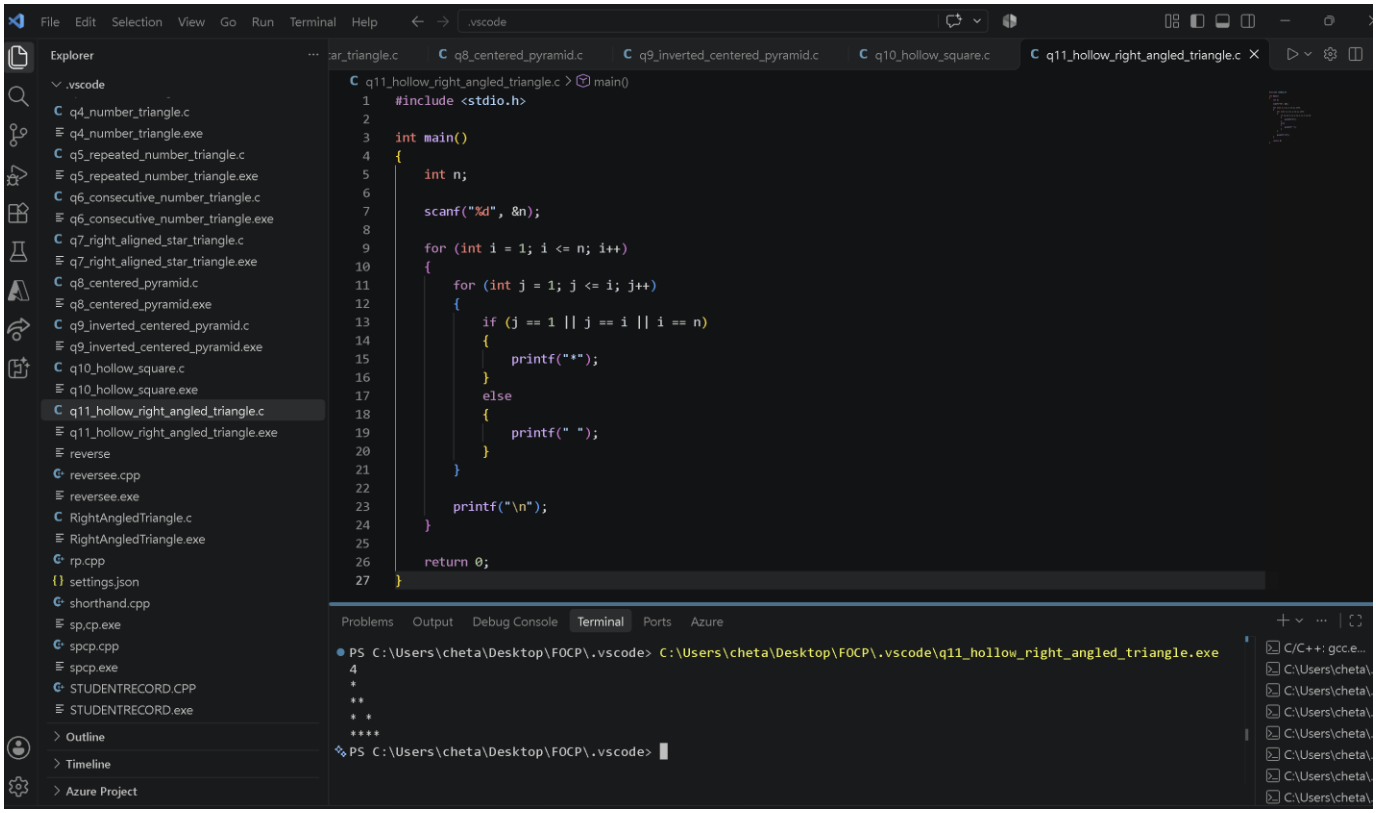
Problem statement: Given n, print the outline of a left-aligned right-angled triangle. Print asterisks in the first column, on the diagonal boundary, and across the final row. Print spaces inside.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Preserve interior spaces; the final row contains n asterisks.

Sample Input 5	**Sample Output** * ** * * * * *****

Hint: A position is on the outline if it is in the first column, on the diagonal, or in the final row.



Question 12 — Print a Multiplication Table Grid

Difficulty: Moderate

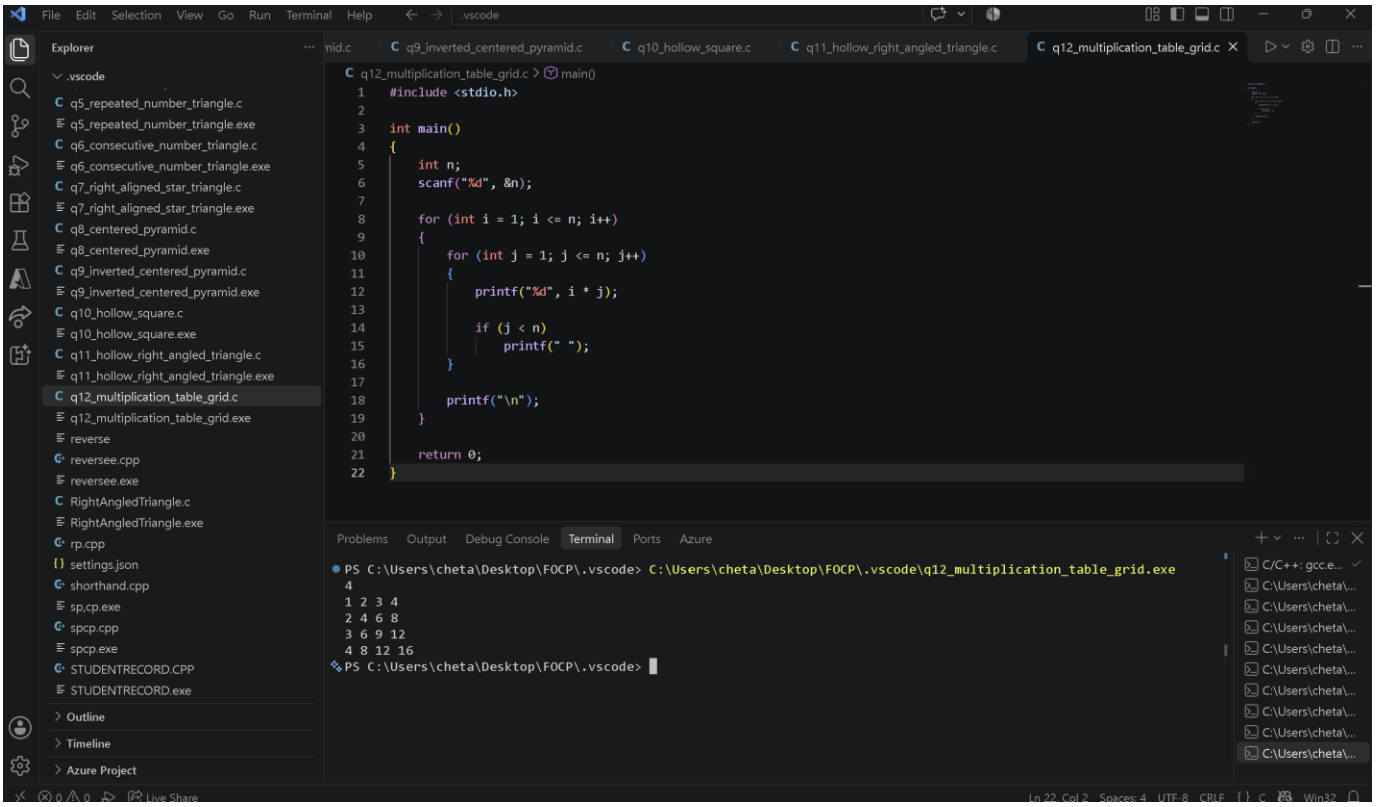
Problem statement: Given n, print an n-by-n multiplication grid. At row i and column j, print i multiplied by j. Separate values with one space.

Input format: One integer n (1 ≤ n ≤ 12).

Output format: Print n lines, each containing n products separated by one space.

Sample Input 3	**Sample Output** 1 2 3 2 4 6 3 6 9

Hint: The value at each position is the row number multiplied by the column number.



Question 13 — Print a Checkerboard

Difficulty: Moderate

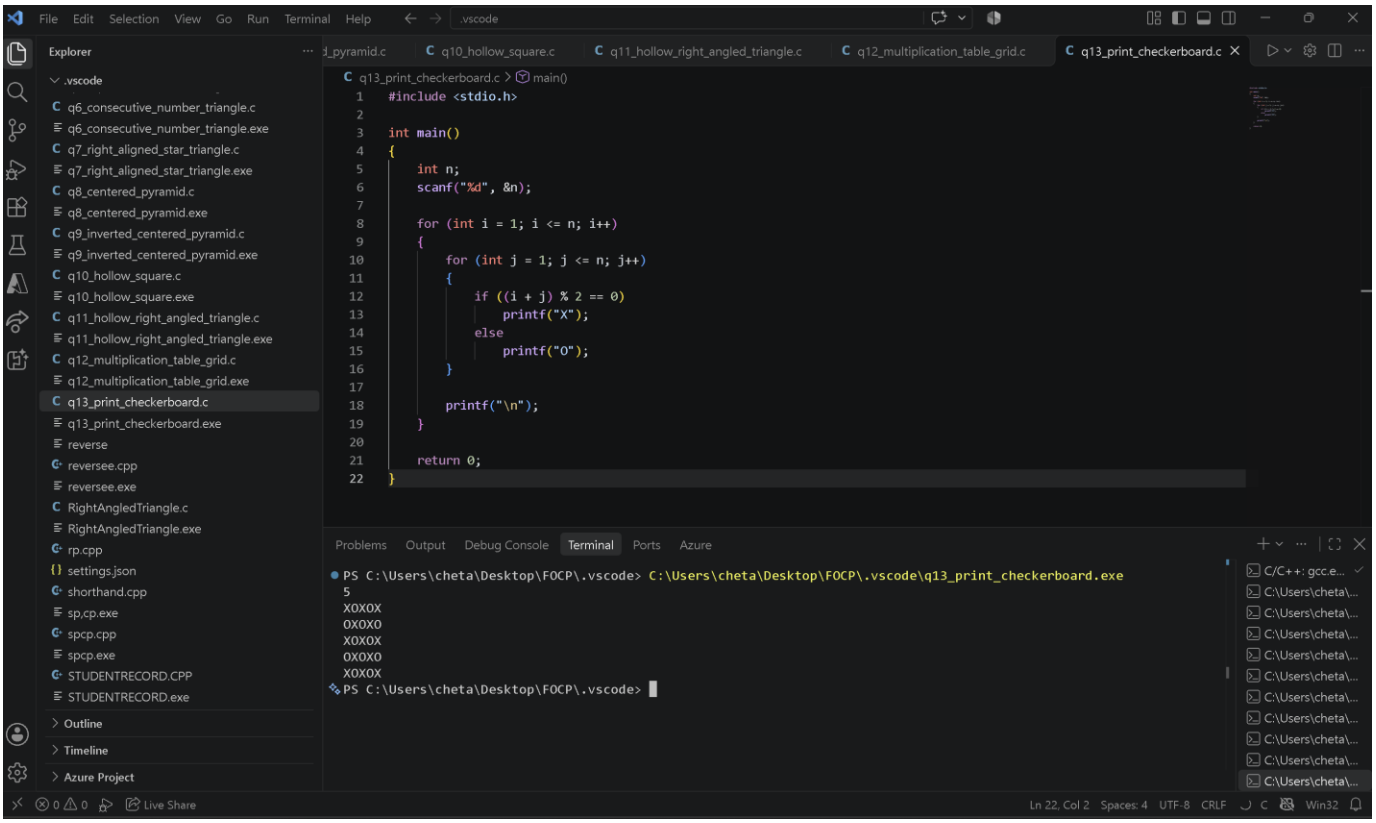
Problem statement: Given n, print an n-by-n checkerboard using X and O. The top-left character must be X; adjacent horizontal and vertical cells alternate.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of n characters, without spaces between characters.

Sample Input 4	**Sample Output** X0X0 0X0X X0X0 0X0X

Hint: Use whether the sum of the row and column indices is even or odd.



Question 14 — Print Floyd's Triangle

Difficulty: Easy–Moderate

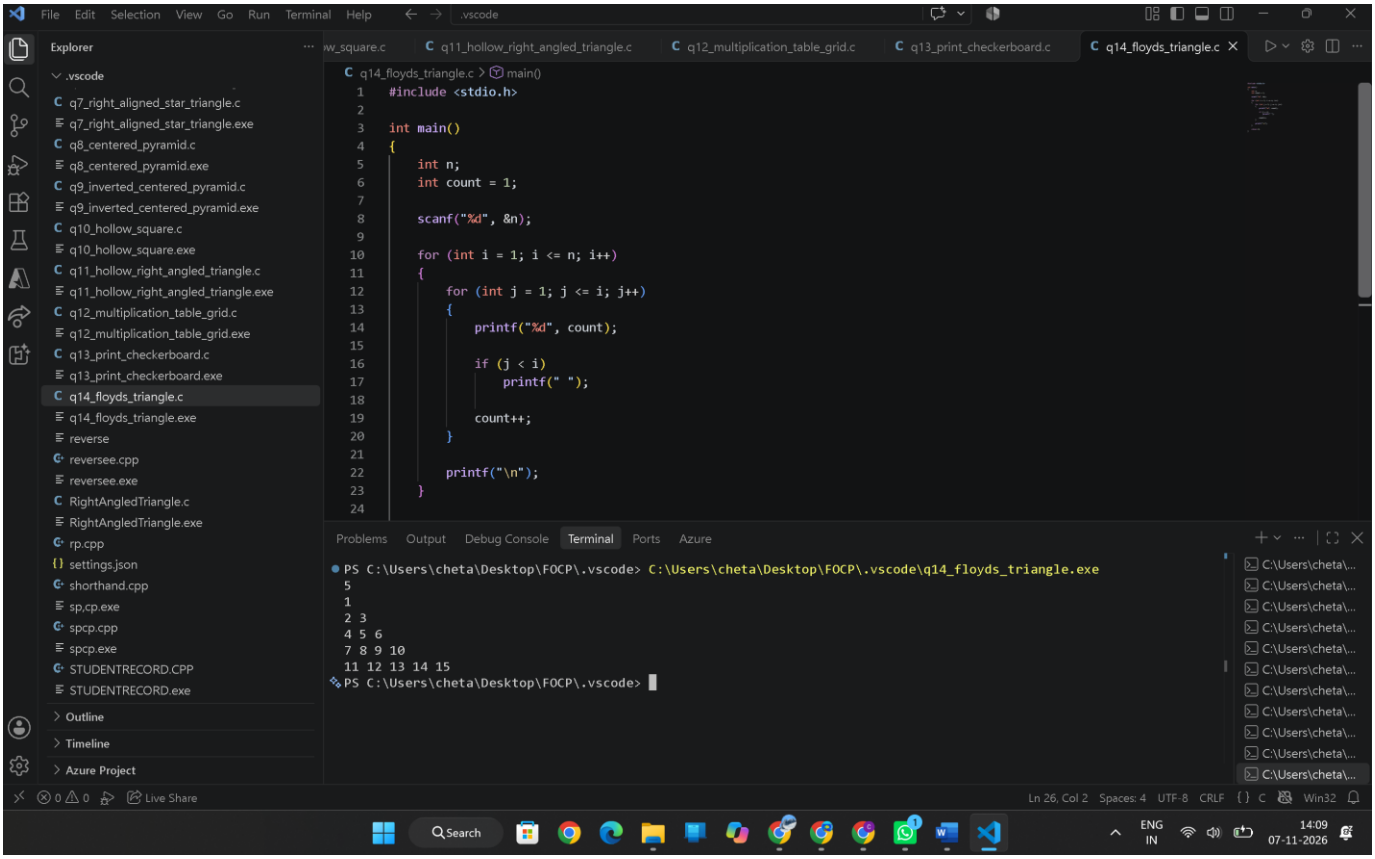
Problem statement: Given n, print Floyd's triangle with n rows. Fill it with consecutive positive integers starting at 1, placing one number on row 1, two on row 2, and so on.

Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines with numbers separated by one space.

Sample Input 4	**Sample Output** 1 2 3 4 5 6 7 8 9 10

Hint: Use a counter that continues increasing across rows.



Question 15 — Print a Butterfly Pattern

Difficulty: Moderate

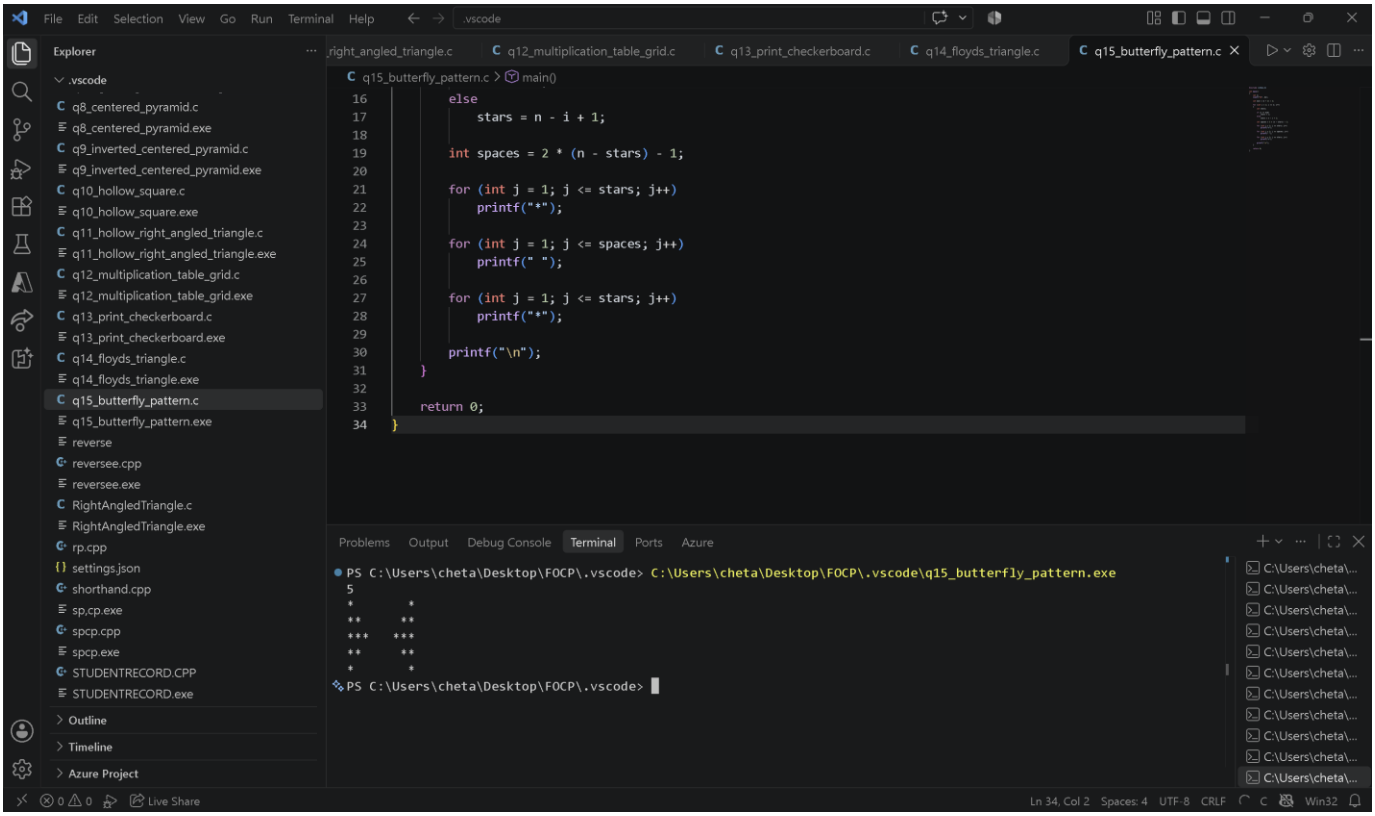
Problem statement: Given an odd integer n, print a butterfly made of asterisks. The top half widens row by row and the lower half mirrors it. Use spaces between the left and right wings.

Input format: One odd integer n (3 ≤ n ≤ 19).

Output format: Print n lines. Each line contains left and right wings with a central gap.

Sample Input 5	**Sample Output** * * ** ** *** *** ** ** * *

Hint: Build the upper half with increasing wing sizes, then mirror it.



Question 16 — Print a Diamond Pattern

Difficulty: Moderate

Problem statement: Given n, print a centered diamond. The upper half has n rows including the widest row; then print n - 1 rows in decreasing width.

Input format: One integer n (1 ≤ n ≤ 10), representing the number of rows in the upper half.

Output format: Print 2n - 1 centered lines.

Sample Input 3	**Sample Output** * *** ***** *** *

Hint: Print an increasing pyramid, then a decreasing half without repeating the widest row.

